

Niayesh Afshordi

Astrophysics, cosmology, and gravity

Professor, Department of Physics and Astronomy, University of Waterloo

Associate Faculty, Perimeter Institute for Theoretical Physics

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Comprehensive public CV — updated July 2026

Research profile

Niayesh Afshordi is an astrophysicist working at the interface of cosmology, gravitation, quantum theory, and observation. His research program connects deep questions in fundamental physics to gravitational-wave data, the cosmic microwave background, large-scale structure, compact objects, and other astrophysical probes. Current themes include quantum aspects of black holes, gravitational-wave echoes and strong-gravity tests, early-universe cosmology, dark matter and dark energy, missing baryons, and the observational consequences of modified gravity.

Academic appointments

Full Professor, Department of Physics and Astronomy, University of Waterloo	2022–present
Professor in the Astrophysics and Gravitation group; founding member of the Waterloo Centre for Astrophysics (2018).	
Research Associate Faculty, Perimeter Institute for Theoretical Physics	2009–present
Associate Faculty in the Cosmology and Gravitation group. Associate Professor, University of Waterloo	
Assistant Professor, University of Waterloo	2009–2015
Distinguished Research Fellow, Perimeter Institute	2007–2009
Postdoctoral Fellow, Institute for Theory and Computation, Harvard–Smithsonian Center for Astrophysics	2004–2007

Education

PhD, Astrophysics, Princeton University	2000–2004
Thesis: <i>The Other 99 Percent.</i>	
BSc, Physics, Sharif University of Technology	1996–1999

Selected honors

Buchalter Cosmology Prize	2019; 2015
Professor M. K. Vainu Bappu Gold Medal, Astronomical Society of India	2011
Ontario Early Researcher Award	2011
NSERC Discovery Accelerator Supplement	2010–2013

Selected research support

NSERC Discovery Grant: “Quantum Black Holes, Vacuum, and the Big Bang: Unraveling cosmic mysteries with Multi-Messenger Astronomy”	2025–2030
Perimeter Travel and Technology Fund	2009–2030
University of Waterloo Interdisciplinary Trailblazer Grant: “Quantum Spacetime in the Cosmos: From Conception to Reality”	2022–2025

Selected supervision

- Alice Chen (PhD): machine learning and cosmic structure formation.
- Conner Dailey (PhD): the initial-boundary-value problem in numerical relativity.
- Ruolin Liu (MSc): quadratic gravity in the Big Bang and black holes.
- Nabeel Rasheed (undergraduate): analytic models for gravitational-wave form templates; subsequently PhD student at MIT (recorded in the 2025 CCV).
- Jolene Zheng (undergraduate): COVID forecast modelling and online dashboard; subsequently Software Engineer at Bloomberg (recorded in the 2025 CCV).
- Hannah Dykaar (undergraduate): gravitational-wave echoes; subsequently PhD student at the University of Toronto (recorded in the 2025 CCV).

Current roles and destinations are reviewed before each public CV update.

Selected recent publications

- A. Y. Chen, P. Halper, and N. Afshordi, “Copenhagen Survey on Black Holes and Fundamental Physics,” *Studies in History and Philosophy of Physics* (2025), submitted; arXiv:2503.15776.
- J. Yang, N. Afshordi, M. Torabian, S. A. Jafari, and G. Baskaran, “Smart Holes: Analogue black holes with the right temperature and entropy,” *JHEP* 07 (2025) 226.
- M. Mylova and N. Afshordi, “Effective cuscuton theory,” *JHEP* 04 (2024) 1–20.
- R. Y. Wen, L. T. Hergt, N. Afshordi, and D. Scott, “A cosmic glitch in gravity,” *JCAP* 03 (2024) 045.
- Y. Li, R. Liu, C. Dailey, and N. Afshordi, “Detecting cosmological scalar fields using orbital networks of quantum sensors,” *Physical Review D* 110 (2024) 043028.
- A. Y. Chen and N. Afshordi, “Outskirts of dark matter haloes,” *Physical Review D* 107 (2023) 103526.
- J. N. Borissova, A. Held, and N. Afshordi, “Scale-invariance at the core of quantum black holes,” *Classical and Quantum Gravity* 40 (2023) 075011.
- N. Oshita and N. Afshordi, “From entropy to echoes: Counting the quasi-normal modes and the quantum limit of silence,” *Physics Letters B* 841 (2023) 137901.
- N. Afshordi and J. Magueijo, “Lower bound on the cosmological constant from the classicality of the early Universe,” *Physical Review D* 106 (2022) 123518.
- T. Yang, M. J. Hudson, and N. Afshordi, “A universal profile for stacked filaments from cold dark matter simulations,” *MNRAS* 516 (2022) 6041–6054.

For the complete and current publication record, see [arXiv](#), [INSPIRE](#), [NASA ADS](#), or [Google Scholar](#).

Public engagement and talks

Afshordi gives research seminars, public lectures, and writing on cosmology, gravity, and how science works. A regularly updated catalogue of talk materials and public work is available at nafshordi.com/talks.

Complete professional record

This appendix is generated from the 15 August 2025 Canadian Common CV. The protected cover page and all personal contact, demographic, citizenship, and identifier fields are excluded. It is retained here to make the public academic CV comprehensive; facts marked as current in the source are rechecked before each update. 3 source line(s) containing protected personal fields were redacted during generation.

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Language Skills

Language Read Write Speak Understand Peer Review

English Yes Yes Yes Yes Yes

Persian Yes Yes Yes Yes Yes

Degrees

2000/9 - 2004/11 Doctorate, PhD, Astrophysics, Princeton University

Degree Status: Completed

Thesis Title: The Other 99 Percent

Transferred to PhD without completing Masters?: Yes

Supervisors: Spergel, David, 2000/9 - 2004/11

Research Disciplines: Astronomy and Astrophysics

1996/9 - 1999/5 Bachelor's Honours, BSc, Physics, Sharif University of Technology

Degree Status: Completed

Recognitions

2020/1 The Buchalter Cosmology Prize 2019

Dr. Ari Buchalter

Prize / Award

The \$10,000 First Prize was awarded to Dr. Jahed Abedi of the Max-Planck-Institut für Gravitationsphysik (Albert-Einstein-Institut) and Leibniz Universität Hannover and Dr. Niayesh Afshordi of the University of Waterloo and the Perimeter Institute for Theoretical Physics, for their work entitled "Echoes from the Abyss: A highly spinning black hole remnant for the binary neutron star merger GW170817", published in the Journal of Cosmology and Astroparticle Physics and recognized by the judging panel as “a bold and innovative step towards understanding quantum gravitational phenomena, using observational data to demonstrate gravitational wave echoes from a binary neutron star merger, which could indicate that black hole interiors may be more complex than the simple regions predicted by General Relativity.”

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2016/1 The Buchalter Cosmology Prize 2015

Dr. Ari Buchalter

Prize / Award

The \$2,500 Third Prize was awarded to Dr. Niayesh Afshordi of the University of Waterloo and the Perimeter Institute for Theoretical Physics and Dr. Elliot Nelson of Perimeter Institute, for their work entitled “Cosmological Non-Constant Problem: Cosmological bounds on TeV-scale physics and beyond”, published in the Journal of Physical Review D and recognized by the judging panel as “an intriguing proposition that the Planck scale of quantum gravity may soon be accessible by particle accelerators, based on the expected white noise fluctuations in the cosmological constant, which might not be constant at all but vary widely in space.”

2011/10 Professor M.K.Vainu Bappu Gold Medal

Astronomical Society of India

Prize / Award

The Vainu Bappu Memorial Gold Medal honours outstanding astronomers worldwide under the age of 35. I was one of the two recipients of this honour in 2011.

2011/9 Early Researcher Award - 150,000 (Canadian dollar)

Government of Ontario

Prize / Award

The ERA program helps promising, recently-appointed Ontario researchers build their research teams of undergraduates, graduate students, post-doctoral fellows, research assistants, associates and technicians. The goal of the program is to improve Ontario’s ability to attract and retain the best and brightest research talent.

2010/9 - 2013/9 Discovery Accelerator Supplement - 120,000

Natural Sciences and Engineering Research Council of Canada (NSERC)

Prize / Award

The Discovery Accelerator Supplement award provides substantial and timely resources to a small group of researchers whose research proposals suggest and explore high-risk, novel or potentially transformative concepts and lines of inquiry, and are likely to have impact by contributing to groundbreaking advances in the area. I was one of the 8 researchers in Physics who received this award in 2010.

User Profile

Researcher Status: Researcher

Research Career Start Date: 1998/01/01

Engaged in Clinical Research?: No

Key Theory / Methodology: General Relativity, Quantum Mechanics, Astrophysical and Cosmological Observations, Numerical Simulations, Large Data, Statistical Methods, Theoretical Modeling

Research Interests: I lead a comprehensive and ambitious research program, which spans a wide area, from Astrophysics/Cosmology to Fundamental Physics, and includes a diverse spectrum of observational, numerical, and analytic elements. The program aims at linking deep mysteries in fundamental physics to novel observational results and techniques, which can be used to shed light on the structure and laws of our Universe. The areas that my group and I currently focus on are: 1. Phenomenology of Dark Energy and cosmological modifications of General Relativity 2. Distribution of Dark Matter on small scales and novel windows for Dark Matter detection 3. Observational consequences of Aether theories of gravity, and their connection to Quantum Gravity and Cosmological Constant Problem 4. Early Universe, and its links to foundations of Gravity and Quantum Mechanics 5. Missing baryons, and observational probes of intergalactic plasma 6. Physics of Neutron Stars, Black Holes

and Accretion Disks.

Research Experience Summary: I am an Astronomer, and a Physicist; in short I am an Astrophysicist. Currently, I am a Professor in the Astrophysics and Gravitation group at the Department of Physics and Astronomy, University

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of Waterloo. I am also an Research Associate Faculty in the Cosmology and Gravitation group at the Perimeter Institute for Theoretical Physics (PI), the premier Canadian institute for research in theoretical Physics. I finished my PhD at Princeton University Observatory in August 2004. I was then a fellow at the Institute for Theory and Computation (ITC) at the Harvard-Smithsonian Center for Astrophysics (2004-2007), and a Distinguished Research Fellow at PI (2008-2009). To date, I have published more than 120 peer-reviewed articles, which have been cited more than 7600 times. I have also (co-)supervised 86 students, and 28 postdoctoral fellows. Moreover, I have co-organized 31 international scientific meetings/seminar series, and raised over 1.5 million dollars in external funding.

Research Specialization Keywords: Astrophysics, Cosmology, Gravitational Waves, Physics of Gravity, Cosmic Microwave Background, Dark Matter, Dark Energy, Black Hole Physics and Astrophysics, Missing Baryons, Lorentz Violation, Quantum Gravity, COVID-19 epidemiology, Flash Radiotherapy

Disciplines Trained In: Astronomy and Astrophysics, Physics

Research Disciplines: Astronomy and Astrophysics, Physics, Epidemiology and Biostatistics, Radiology

Areas of Research: Galactic and Cosmic, Universe Structure, Gravitation, Quantum Phenomena, Elementary Particles

Fields of Application: Foundations and Knowledge Acquisition, Biomedical Aspects of Human Health, Public Health

Employment

2022/9 Full Professor of Physics and Astronomy

Physics and Astronomy, University of Waterloo

Full-time, Professor

Tenure Status: Tenure

Professor in the Astrophysics and Gravitation group. Founding Member of the Waterloo Centre for Astrophysics (2018)

2009/9 Research Associate Faculty

Cosmology and Gravitation, Waterloo, Perimeter Institute for Theoretical Physics

Full-time

Tenure Status: Non Tenure Track

A Renewable appointment as a Perimeter Institute for Theoretical Physics Faculty Member.

2015/9 - 2022/7 Associate Professor

Physics and Astronomy, Waterloo Centre for Astrophysics, University of Waterloo

Full-time, Associate Professor

Tenure Status: Tenure

Associate Professor in the Astrophysics and Gravitation group. Founding Member of the Waterloo Centre for Astrophysics (2018)

2009/9 - 2015/9 Assistant Professor

Physics and Astronomy, Science, University of Waterloo

Full-time, Assistant Professor

Tenure Status: Tenure Track

Assistant Professor in the Astrophysics and Gravitation group

2012/9 - 2013/8 Visiting Professor

Physics, North, SUNY at Buffalo

Part-time, Visiting Professorship

Tenure Status: Non Tenure Track

I visited this department on a regular basis and engaged and collaborated with the faculty

in Cosmology.

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2007/9 - 2009/8 Distinguished Research Fellow

Cosmology and Gravitation, Waterloo, Perimeter Institute for Theoretical Physics

Full-time

Tenure Status: Non Tenure Track

2004/9 - 2007/8 Postdoctoral Fellow

Astrophysics, Institute for Theory and Computation, Harvard-Smithsonian Center for

Astrophysics

Full-time

Tenure Status: Non Tenure Track

2000/9 - 2004/8 Graduate Student Instructor and Researcher

Astrophysical Sciences, Arts and Sciences, Princeton University

Part-time

Tenure Status: Non Tenure Track

1999/9 - 2000/8 Graduate Student Teaching Assistant

Physics, Arts and Sciences, Brown University

Part-time

Tenure Status: Non Tenure Track

1998/9 - 1999/8 Research Associate

Physics, Tehran, Institute for Theoretical Studies in Physics and Mathematics (IPM)

Full-time

Tenure Status: Non Tenure Track

1996/1 - 1998/8 Instructor

Iranian National Physics Olympiad Camp, International Physics Olympiad

Affiliations

The primary affiliation is denoted by (*)

(*) 2015/9 Associate Professor, University of Waterloo

2009/9 Associate Faculty, Perimeter Institute for Theoretical Physics

Leaves of Absence and Impact on Research

2019/9 - 2020/8 Sabbatical, University of Waterloo

I was on sabbatical at Perimeter Institute, and did not have any research/teaching duties at U-Waterloo.

Research Funding History

Awarded [n=4]

2009/9 - 2030/7 Perimeter Travel and Technology Fund (ongoing), Grant

Principal Investigator Funding Sources:

Perimeter Institute

Travel and Technology Fund

Total Funding - 210,000

Portion of Funding Received - 210,000

Funding Competitive?: No

2025/4 - 2030/3 Quantum Black Holes, Vacuum, and the Big Bang: Unraveling cosmic mysteries with

Principal Investigator Multi-Messenger Astronomy, Grant

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Funding Sources:

2025/4 - 2030/3 Natural Sciences and Engineering Research Council of Canada
(NSERC)

Discovery Grant

Total Funding - 250,000 (Canadian dollar)

Portion of Funding Received - 0

Funding Renewable?: Yes

Funding Competitive?: Yes

2022/9 - 2025/8 Quantum Spacetime in the Cosmos: From Conception to Reality, Grant

Principal Investigator Funding Sources:

University of Waterloo

Interdisciplinary Trailblazer grant

Total Funding - 80,000

Portion of Funding Received - 68,000

Funding Competitive?: Yes

Co-investigator : Fraser, Doreen;

Collaborator : Weinstein, Steven

2020/4 - 2025/3 Novel Astrophysical Probes of Fundamental Physics, Grant

Principal Applicant Funding Sources:

Natural Sciences and Engineering Research Council of Canada
(NSERC)

Discovery

Total Funding - 175,440

Portion of Funding Received - 175,440

Funding Competitive?: Yes

Completed [n=5]

2021/9 - 2023/8 HQP Pooled Resources Competition, Grant

Principal Investigator Funding Sources:

Arthur B. McDonald Canadian Astroparticle Physics Research
Institute

HQP Pooled Resources Competition

Total Funding - 40,000

Portion of Funding Received - 40,000

Funding Competitive?: Yes

2015/4 - 2020/3 Novel Astrophysical Probes of Fundamental Physics, Grant

Principal Investigator Funding Sources:

Natural Sciences and Engineering Research Council of Canada
(NSERC)

Discovery

Total Funding - 210,000

Portion of Funding Received - 210,000

Funding Competitive?: Yes

2011/4 - 2016/3 Early Researcher Award, Grant

Principal Applicant

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Funding Sources:

2011/9 - 2016/9 Ontario Ministry of Research and Innovation (ON)

Early Researcher Award

Total Funding - 150,000

Portion of Funding Received - 150,000

Funding Competitive?: Yes

2010/4 - 2015/3 Discovery Grant, Grant

Principal Applicant Funding Sources:

2010/4 - 2015/3 Natural Sciences and Engineering Research Council of Canada (NSERC)

Total Funding - 280,000

Portion of Funding Received - 280,000

Funding Competitive?: Yes

2010/4 - 2013/3 Discovery Accelerator Supplement, Grant

Principal Applicant Funding Sources:

2010/4 - 2013/3 Natural Sciences and Engineering Research Council of Canada (NSERC)

Discovery Accelerator Supplement

Total Funding - 120,000

Portion of Funding Received - 120,000

Funding Competitive?: Yes

Courses Taught

Instructor, Science, University of Waterloo

Course Title: Introductory Astronomy

Course Code: 238

Course Level: Undergraduate

Academic Session: Winter

Number of Students: 130

Lecture Hours Per Week: 3

Student/Postdoctoral Supervision

Bachelor's [n=8]

2023/9 - 2025/8 Rasheed, Nabeel (Completed) , University of Waterloo

Principal Supervisor Thesis/Project Title: NSERC USRA project: "Analytic models for gravitational wave form templates"

Present Position: PhD Student, Massachusetts Institute of Technology

2023/9 - 2024/7 Yin, Xitai (John) (Completed) , University of Waterloo

Principal Supervisor Thesis/Project Title: Black Holes in Quantum Gravity

Present Position: Master's student in Physics, University of Waterloo

2023/9 - 2025/8 Tang, Ruijing (In Progress) , University of Waterloo

Principal Supervisor Thesis/Project Title: BSc thesis project: "Exploring Gravitational Effects on accretion disk dynamics in the Teleparallel Born-Infeld gravity"

Present Position: BSc Student, University of Waterloo

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2023/8 - 2021/9 Liu, Ruolin (Completed) , University of Waterloo
Principal Supervisor Thesis/Project Title: BSc Thesis Project: "Detecting cosmological scalar fields using orbital networks of quantum sensors"
Present Position: MSc Student, University of Waterloo

2022/1 - 2025/4 Subhi, Hassan (In Progress) , University of Waterloo
Principal Supervisor Student Degree Expected Date: 2025/4
Thesis/Project Title: Research Project: "Gravitational Wave Echoes in Pulsar Timing Arrays"
Present Position: Undergraduate Student, University of Waterloo

2021/2 - 2021/9 Zheng, Jolene (Completed) , University of Waterloo
Principal Supervisor Thesis/Project Title: Covid forecast modeling and online dashboard
Present Position: Software Engineer, Bloomberg

2021/2 - 2021/9 Olalekan Koiki, Shafika (Completed) , University of Waterloo
Principal Supervisor Thesis/Project Title: Co-op project: "Covid forecast modeling and online dashboard"
Present Position: none

2021/1 - 2021/8 Wisniewski, Kasia (Completed) , University of Waterloo
Principal Supervisor Thesis/Project Title: BSc thesis project: "Searching for Black Hole Echoes from the Failed Supernova AT2017be"
Present Position: Operations Technician, GHGSat Inc
Bachelor's Honours [n=20]

2024/9 - 2025/5 Shah, Saral (In Progress) , University of Waterloo
Principal Supervisor Student Degree Expected Date: 2025/5
Thesis/Project Title: Bachelor's thesis: "Dynamical Friction in a neutrinos sea"
Present Position: Undergraduate Student, University of Waterloo

2024/9 - 2025/5 Yanover, Lyvia (In Progress) , University of Waterloo
Principal Supervisor Student Degree Expected Date: 2025/5
Thesis/Project Title: Bachelor's thesis: "Bi-Thermal Big Bang"
Present Position: Undergraduate Student, University of Waterloo

2024/9 - 2025/5 Murali, Nikhil (In Progress) , University of Waterloo
Principal Supervisor Student Degree Expected Date: 2025/5
Thesis/Project Title: Bachelor's Thesis: "Quantum Black Holes: quantum corrections at the tip of a cigar"
Present Position: Undergraduate Student, University of Waterloo

2024/9 - 2025/5 Brisebois, Alex Dominic Raul (In Progress) , University of Waterloo
Principal Supervisor Student Degree Expected Date: 2025/5
Thesis/Project Title: Bachelor's thesis: "Clustering in the Action-Angle Space"
Present Position: Undergraduate Student, University of Waterloo

2023/9 - 2024/1 Zhao, Guandi (Completed) , Fudan University, China
Principal Supervisor Thesis/Project Title: BSc visiting student at Perimeter Institute for Theoretical Physics: "Thermal SZ contribution from Low mass galaxies in the unWISE survey"
Present Position: MSc student, ETH Zurich, Switzerland

2023/9 - 2024/8 Goel, Lakshay (Completed) , University of Waterloo
Co-Supervisor Thesis/Project Title: Accelerated detector in a superposed spacetime
Present Position: none

2023/9 - 2024/7 Gifford, Hazel (Completed) , University of Waterloo
Principal Supervisor Thesis/Project Title: Bohmian Rhapsody: Cuscuton and Cosmology

Present Position: MSc Student, Brock University

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2022/9 - 2023/6 Schiebelbein, Aryanna (Completed) , University of Waterloo

Principal Supervisor Thesis/Project Title: BSc Thesis Projects: "Mystery noise in LIGO"

Present Position: PhD Student, University of Toronto

2022/1 - 2024/7 Li, Yu (Completed) , University of Waterloo

Principal Supervisor Thesis/Project Title: NSERC USRA project: "Detecting cosmological scalar fields using orbital networks of quantum sensors"

Present Position: PhD Student, California Institute of Technology

2021/5 - 2023/8 Lawrence, Jensen Pierce Thomas (Completed) , University of Waterloo

Principal Supervisor Thesis/Project Title: Research Project: "Universal horizon formation in the incompressible limit of Einstein-Aether theory"

Present Position: Department of Earth, Atmospheric and Planetary Sciences,
Massachusetts Institute of Technology, PhD Student

2020/9 - 2023/8 Wen, Robin (Completed) , Department of Applied Math, University of Waterloo

Principal Supervisor Thesis/Project Title: BSc Thesis project: "A Cosmic Glitch in gravity"

Present Position: PhD Student, California Institute of Technology

2020/9 - 2021/8 Stewart, Meagan (Completed) , University of Waterloo

Principal Supervisor Thesis/Project Title: BSc thesis project: "Inferring Transient Weak Lensing from Quasar Light Curves"

Present Position: MSc Student, Simon Fraser University, British Columbia

2020/5 - 2022/5 Shawqi, Shafayat (Completed) , University of Waterloo

Principal Supervisor Thesis/Project Title: BSc Thesis Project: "Sorkin-Johnston Effective Field Theory"

Present Position: PhD Student, University of Alberta

2020/1 - 2020/5 Duff, Petra Mary (Completed) , University of Waterloo

Principal Supervisor Thesis/Project Title: NSERC USRA project: "Echoes from the Abyss"

Present Position: PhD Student, University of Toronto

2019/9 - 2022/8 Minhas, Palka (Completed) , University of Waterloo

Principal Supervisor Thesis/Project Title: Fast Radio Bursts

Present Position: MSc Student, McGill University

2017/1 - 2017/4 Adair, Connor (Completed) , University of Waterloo

Principal Supervisor Thesis/Project Title: NSERC USRA recipient working on Gravitational Wave Echoes

Present Position: MSc Student, University of British Columbia

2016/1 - 2016/4 Dykaar, Hannah (Completed) , University of Waterloo

Principal Supervisor Thesis/Project Title: NSERC USRA research project: "Echoes from the Abyss"

Present Position: PhD Student, University of Toronto

2016/1 - 2016/4 Elgood, Zachary (Completed) , University of Waterloo

Principal Supervisor Thesis/Project Title: NSERC USRA recipient working on Einstein-Aether Black Holes

Present Position: La Caixa Doctoral Scholar, Institute of Theoretical Physics (IFT), Madrid,
Spain

2010/9 - 2011/6 Bovard, Luke (Completed) , University of Waterloo

Co-Supervisor Thesis/Project Title: Undergraduate B.Sc. Honours Research Project

Present Position: Postdoctoral Researcher, Goethe University, Germany

2008/9 - 2009/8 Mueller, Matthias (Completed) , Ulm University

Co-Supervisor Thesis/Project Title: Undergraduate Research Project

Present Position: Postdoctoral Fellow, University of Florence

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Master's non-Thesis [n=7]

2023/9 - 2025/8 Liu, Ruolin (In Progress) , University of Waterloo

Principal Supervisor Thesis/Project Title: Quadratic Gravity in Big Bang and Black Holes

Present Position: MSc Student, University of Waterloo

2023/3 - 2023/12 Friedman-Shaw, Batia (Completed) , Perimeter Institute for Theoretical Physics/ University

Principal Supervisor of Waterloo

Thesis/Project Title: MSc Project:"Doppler Bias: Impact of Peculiar Velocities on Color

Selection and the Large Scale Structure of Galaxy Surveys"

Present Position: PhD Student, Perimeter Institute for Theoretical Physics/ University of

Waterloo

2022/9 - 2023/6 Hrytseniak, Oleksandra (Completed) , Perimeter Institute for Theoretical Physics/

Co-Supervisor University of Waterloo

Thesis/Project Title: Project supervision: "Search for flattened non-gaussianity"

Present Position: PhD student, Perimeter Institute for Theoretical Physics/ University of

Waterloo

2019/9 - 2020/8 Chua, Wan Zen (Completed) , University of Waterloo

Principal Supervisor Thesis/Project Title: PSI Research Project: "Electromagnetic Albedo of Fuzzballs"

Present Position: PhD student, Cornell University

2015/9 - 2016/6 Wang, Qingwen (Completed) , Perimeter Institute for Theoretical Physics

Principal Supervisor Thesis/Project Title: Fuzzball approach to black holes

Present Position: PhD Student, University of Waterloo

2013/8 - 2014/6 Roberts, Brodyn (Completed) , University of Waterloo/Perimeter Institute for Theoretical

Principal Supervisor Physics

Thesis/Project Title: PSI Research Project

Present Position: Software Engineer, Index Exchange, Toronto, Canada

2010/9 - 2012/6 Agarwal, Abhineet (Completed) , University of Waterloo

Principal Supervisor Thesis/Project Title: Thermal Tachyacoustic Cosmology

Present Position: PhD Student, Jamia Millia University, India

Master's Thesis [n=12]

2022/9 - 2024/8 Yang, Jiayue (Completed) , University of Waterloo

Co-Supervisor Thesis/Project Title: Entropy of Analogue Black Holes in Tilted Dirac Cone Materials

Present Position: MSc Student, University of Waterloo

2021/1 - 2021/12 Golafshani, Hamid (Withdrawn) , University of Waterloo

Principal Supervisor Thesis/Project Title: Topics in Astrophysics and Cosmology

Present Position: Professor of Mechanical Engineering, Conestoga College, Waterloo, ON

2018/9 - 2020/8 Hergott, Samantha (Completed) , University of Waterloo

Principal Supervisor Thesis/Project Title: Unifying Big Bang and Black Holes

Present Position: PhD Student, York University

2018/9 - 2019/8 Yang, Tianyi (Completed) , University of Waterloo

Principal Supervisor Thesis/Project Title: Cosmic Filaments, Galactic Kinematics in Gaia

Present Position: PhD Student, University of Edinburgh

2018/9 - 2020/8 Chen, Alice (Completed) , University of Waterloo

Principal Supervisor Thesis/Project Title: Cosmological Structure Formation

Present Position: PhD Student, University of Waterloo

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2017/9 - 2018/7 Boruah, Supranta (Completed) , University of Waterloo

Co-Supervisor Thesis/Project Title: Constraining the mass of Milky Way based on clustering in the action space of Gaia DR2 stars

Present Position: Postdoctoral Fellow, University of Pennsylvania

2016/9 - 2017/8 Kim, Hyungjin (Completed) , University of Waterloo

Principal Supervisor Thesis/Project Title: The Early Universe: New aspects of Cuscuton Theory, Modified Dispersion Relations and Pulsar Constraints on Vacuum Noise

Present Position: PhD Student, University of Waterloo

2014/9 - 2016/8 Meiers, Michael (Completed) , University of Waterloo

Co-Supervisor Thesis/Project Title: Universal Horizons in Black Holes

Present Position: PhD Student, University of California-Davis

2012/8 - 2013/6 Okoli, Chiamaka (Completed) , University of Waterloo/Perimeter Institute for Theoretical Physics

Thesis/Project Title: Concentration of cosmological haloes

Present Position: Deceased

2011/9 - 2013/5 Yazdi, Yasaman (Completed) , University of Waterloo

Principal Supervisor Thesis/Project Title: A Spacetime Approach to Defining Vacuum States and Entropy

Present Position: Avadh Bhatia Postdoctoral Fellow, University of Alberta

2011/8 - 2012/6 Schuhmann, Robert (Completed) , University of Waterloo/Perimeter Institute for

Principal Supervisor Theoretical Physics

Thesis/Project Title: PSI Research Project

Present Position: Postdoctoral Research Associate, University of Manchester

2009/8 - 2010/6 Ponce, Yosdanis (Completed) , University of Waterloo/Perimeter Institute for Theoretical

Principal Supervisor Physics

Thesis/Project Title: Cosmology of a Black Hole Lattice

Present Position: Adjunct Faculty in Physics, Miami Dade College, Wolfson Campus

Doctorate [n=39]

2023/10 - 2025/8 Nasiri, Arad (In Progress) , Imperial College London

Co-Supervisor Student Degree Expected Date: 2025/8

Thesis/Project Title: Project Supervision: "Stochastic Dark Matter: Covariant Brownian Motion from Planckian Discreteness (arXiv:2409.02188)"

Present Position: PhD Student, Imperial College London

2023/9 - 2024/12 Patterson, Everett Avison (In Progress) , University of Waterloo

Co-Supervisor Student Degree Expected Date: 2025/12

Thesis/Project Title: Project Supervision: "Accelerated detector in a superposed spacetime (arXiv:2409.12441)"

Present Position: PhD Student, University of Waterloo

2023/1 - 2026/12 Chen, Alice (In Progress) , University of Waterloo and Perimeter Institute for Theoretical

Principal Supervisor Physics

Student Degree Expected Date: 2026/12

Thesis/Project Title: Machine Learning and Cosmic Structure Formation

Present Position: PhD Student

2022/9 - 2025/8 Borissova, Johanna (In Progress) , Perimeter Institute for Theoretical Physics/ University

Co-Supervisor of Waterloo

Thesis/Project Title: Scale-invariance at the core of quantum black holes

Present Position: PhD Student, Perimeter Institute for Theoretical Physics/ University of

Waterloo

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2022/7 - 2024/12 Kim, Leo (In Progress) , Queen's University

Co-Supervisor Thesis/Project Title: Project Supervision: "A Poisson Log-Normal Framework for Dark Sirens"

Present Position: PhD Student, Queen's University

2022/7 - 2022/9 Boyanov Savov, Valentin (Completed) , Universidad Complutense de Madrid

Principal Supervisor Thesis/Project Title: Visiting PhD student: "Aspects of Quantum Black Holes in semi-classical gravity"

Present Position: Postdoctoral Fellow, Instituto Superior Tecnico, Lisbon, Portugal

2020/9 - 2024/12 Dailey, Conner (All But Degree) , University of Waterloo and Perimeter Institute

Principal Supervisor Student Degree Expected Date: 2024/12

Thesis/Project Title: On the Initial Boundary Value Problem in Numerical Relativity

Present Position: PhD student, University of Waterloo and Perimeter Institute

2019/8 - 2020/4 Longo, Luis Felipe (Completed) , University of Sao Paolo, Brazil and Perimeter Institute for

Principal Supervisor Theoretical Physics

Thesis/Project Title: visiting PhD student: "Echo-Diversity in Binary Black Hole Mergers"

Present Position: European Research Council (ERC) Postdoctoral Fellow, Friedrich-Schiller Universitaet, Jena, Germany

2019/2 - 2020/7 Tsuna, Daichi (Completed) , University of Tokyo

Co-Supervisor Thesis/Project Title: visiting PhD student: "Quantum Black Hole Seismology"

Present Position: Sherman Fairchild Postdoctoral Scholar Research Associate, California Institute of Technology

2018/9 - 2019/8 Held, Aaron (Completed) , University of Heidelberg

Principal Supervisor Thesis/Project Title: Visiting PhD student: "Quantum Black Holes in Asymptotic Safety"

Present Position: Postdoctoral Fellow, Ecole Normale Supérieure, Paris, France

2018/9 - 2019/8 Giri, Utkarsh (Completed) , University of Waterloo/Perimeter Institute for Theoretical

Co-Supervisor Physics

Thesis/Project Title: Pressure of haloes in cosmological simulations

Present Position: Postdoctoral Fellow, California Institute of Technology

2018/9 - 2023/8 Saraswat, Krishan (Completed) , University of Waterloo/Perimeter Institute for Theoretical

Principal Supervisor Physics

Thesis/Project Title: Quantum Aspects of Black Holes: From Microstates to Echoes and

Somewhere In-Between

Present Position: Postdoctoral Fellow, University of Southern California

2018/1 - 2018/9 Oshita, Naritaka (Completed) , University of Tokyo

Principal Supervisor Thesis/Project Title: visiting PhD student: "Probing microstructure of black hole spacetimes with gravitational wave echoes"

Present Position: Assistant Professor, Kyoto University, Yukawa Institute, Kyoto, Japan

2017/9 - 2018/4 Dhumuntarao, Aditya (Completed) , University of Minnesota

Principal Supervisor Thesis/Project Title: visiting NSF PhD student: "Temperature for local observers in Quantum Field Theory in curved spacetime"

Present Position: Postdoctoral Researcher, Sandia National Laboratories

2017/9 - 2021/6 Robbins, Matthew (Completed) , University of Waterloo/Perimeter Institute for Theoretical

Principal Supervisor Physics

Thesis/Project Title: Gravitational Waves, Bose-Einstein Condensates, and Modified

Gravity

Present Position: Teaching Faculty (tenure-track), University of Toronto

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2017/9 - 2018/4 Ghersi, Jose Tomas Galvez (Completed) , Simon Fraser University
Principal Supervisor Thesis/Project Title: visiting PhD student: "Temperature for local observers in Quantum Field Theory in curved spacetime"
Present Position: Assistant Professor, Universidad de Ingenieria y Tecnologia, Peru

2016/9 - 2020/6 Wang, Qingwen (Completed) , University of Waterloo/Perimeter Institute for Theoretical Physics
Principal Supervisor Thesis/Project Title: Black Hole Echology
Present Position: Senior Quant, Deloitte Business Consulting and Services

2016/9 - 2018/8 Abedi, Jahed (Completed) , Sharif University of Technology
Co-Supervisor Thesis/Project Title: Black Hole Echoes
Present Position: Postdoctoral Fellow, University of Stavanger, Norway

2016/9 - 2017/8 Haghighi, Mohammad Zhoolideh (Completed) , Sharif University
Principal Supervisor Thesis/Project Title: visiting PhD student: "X-ray Cooling Flows and Modified Gravity"
Present Position: Postdoctoral Fellow, IPM, Tehran

2015/9 - 2016/8 Lynch, Morgan (Completed) , University of Wisconsin-Milwaukee
Principal Supervisor Thesis/Project Title: visiting PhD student: "Temperatures of renormalizable quantum field theories in curved spacetime"
Present Position: Zuckerman Postdoctoral Scholar, Technion

2014/9 - 2018/6 Karami, Mansour (Completed) , University of Waterloo/Perimeter Institute for Theoretical Physics
Co-Supervisor Thesis/Project Title: Dark Matter on Small Scales
Present Position: Senior Machine Learning Engineer, Riskfuel, Toronto

2014/9 - 2018/6 Altamirano, Natacha (Completed) , University of Waterloo/Perimeter Institute for Theoretical Physics
Co-Supervisor Thesis/Project Title: The quantum and the Gravity: Newtonian and Cosmological Applications
Present Position: Director, Data Science, Birch Hill Equity Partners, Toronto, Canada

2014/9 - 2018/6 Okoli, Chiamaka (Completed) , University of Waterloo/Perimeter Institute for Theoretical Physics
Principal Supervisor Thesis/Project Title: Dark Matter and Neutrinos in the Foggy Universe
Present Position: Deceased

2014/4 - 2014/11 Nelson, Elliott (Completed) , Penn State University
Principal Supervisor Thesis/Project Title: Cosmological Non-Constant Problem
Present Position: MSc Student in Computer Science, University of Waterloo, Canada

2013/9 - 2017/6 Yazdi, Yasaman (Completed) , University of Waterloo
Co-Supervisor Thesis/Project Title: Physics and Astrophysics of Black Holes
Present Position: Avadh Bhatia Postdoctoral Fellow, University of Alberta

2013/9 - 2014/8 Narimani, Ali (Completed) , University of British Columbia
Principal Supervisor Thesis/Project Title: visiting PhD student: "How does pressure gravitate?"
Present Position: Data Scientist, Twitter

2013/9 - 2014/6 Karami, Mansour (Completed) , Sharif University of Technology
Co-Supervisor Thesis/Project Title: Cosmic Structure Formation, Modified Gravity and Dark Energy, Gravitational Lensing
Present Position: Manager, Market Risk Measurement, Scotiabank, Canada

2013/9 - 2017/6 Zwane, Nosiphiwo (Completed) , University of Waterloo/Perimeter Institute for Theoretical Physics
Co-Supervisor Thesis/Project Title: Cosmic Structure Formation, Modified Gravity and Dark Energy, Gravitational Lensing

Thesis/Project Title: Cosmological Tests of Causal Set Phenomenology

Present Position: Lecturer, University of Eswatini

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2013/9 - 2014/8 Jeremy Sakstein (Completed) , University of Cambridge

Co-Supervisor Student Degree Start Date: 2010/9

Student Degree Received Date: 2014/7

Thesis/Project Title: visiting PhD student: "Novel Approaches to Modified Gravity"

Present Position: Postdoctoral Fellow, Center for Particle Cosmology, University of Pennsylvania, USA

2011/9 - 2017/6 Gould, Elizabeth (Completed) , University of Waterloo/Perimeter Institute for Theoretical
Principal Supervisor Physics

Thesis/Project Title: A Non-Local Reality: Is there a Phase Uncertainty in Quantum Mechanics?

Present Position: Postdoctoral Fellow, Arthur B. McDonald Canadian Astroparticle Physics Research Institute

2011/9 - 2016/6 Saravani, Mehdi (Completed) , University of Waterloo/Perimeter Institute for Theoretical
Principal Supervisor Physics

Thesis/Project Title: Aspects of Non-locality

Present Position: Corporate Treasury Strategist, Goldman-Sachs, London, UK

2011/9 - 2014/7 Pourhasan, Razieh (Completed) , University of Waterloo/Perimeter Institute for Theoretical
Co-Supervisor Physics

Thesis/Project Title: Adventures in Holography

Present Position: Research Assistant, University of Ottawa, Canada

2011/9 - 2015/5 Kamiab, Farbod (Completed) , University of Waterloo/Perimeter Institute for Theoretical
Principal Supervisor Physics

Thesis/Project Title: Neutron Stars, the Exotica - From Modifying General Relativity to Strong Magnetic Fields

Present Position: Analytics, Facebook, Dublin, Ireland

2011/1 - 2012/8 Ponce, Yosdanis (Withdrawn) , University of Waterloo/Perimeter Institute for Theoretical
Principal Supervisor Physics

Student Degree Start Date: 2011/1

Thesis/Project Title: Black Hole Halo Model/ Holographic Cosmology

Present Position: Adjunct Faculty in Physics, Miami Dade College, Wolfson Campus

2010/9 - 2011/9 Baghram, Shant (Completed) , Sharif University of Technology

Co-Supervisor Thesis/Project Title: The Observational Tests of Dark Energy and Modified Gravity Theories

Present Position: Assistant Professor, Sharif University, Tehran, Iran

2010/9 - 2011/6 Gryb, Sean (Completed) , University of Waterloo/Perimeter Institute for Theoretical
Co-Supervisor Physics

Thesis/Project Title: Shape Dynamics and Mach's Principles: Gravity from Conformal Geometrodynamics

Present Position: Research Associate, Philosophy Department, University of Bristol, England

2010/9 - 2014/6 Aslanbeigi, Siavash (Completed) , University of Waterloo/Perimeter Institute for
Principal Supervisor Theoretical Physics

Thesis/Project Title: Cosmic Atoms: from Causal Sets to Clusters

Present Position: Vice President, Trading Products, BMO Capital Markets, Canada

2008/9 - 2010/6 Prescod-Weinstein, Chanda (Completed) , University of Waterloo

Co-Supervisor Thesis/Project Title: Cosmic Acceleration As Quantum Gravity Phenomenology

Present Position: Assistant Professor, University of New Hampshire

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2007/9 - 2008/8 LoVerde, Marilena (Completed) , Columbia University

Co-Supervisor Thesis/Project Title: Extended Limber Approximation

Present Position: Assistant Professor, Stony Brook University, New York

Post-doctorate [n=27]

2024/3 - 2024/12 Dunskey, David (In Progress) , New York Univesity

Co-Supervisor Thesis/Project Title: Project Supervision: "Dark Matter Disk Accretion into Black Holes"

Present Position: Postdoctoral Fellow, New York Univesity

2024/3 - 2024/12 Ghalsasi, Akshay (In Progress) , Harvard University

Co-Supervisor Thesis/Project Title: Project Supervsion: "Dark Matter Disk Accretion onto Black Holes"

Present Position: Postdoctoral Fellow, Deoartment of Physics, Harvard University

2024/3 - 2026/3 Faraji, Shokoufe (In Progress) , University of Waterloo

Principal Supervisor Thesis/Project Title: Quantum Gravity and Accretion Physics around Black Holes

Present Position: Postdoctoral Fellow, Department of Physics and Astronomy, University of Waterloo

2023/3 - 2024/12 Krolewski, Alex (In Progress) , University of Waterloo/ Perimeter Institute for Theoretical Co-Supervisor Physics

Thesis/Project Title: Impact of Peculiar Velocities on Color Selection and the Large Scale Structure of Galaxy Surveys

Present Position: CITA National Fellow, Waterloo Centre for Astrophysics

2023/1 - 2025/9 Quintin, Jerome (In Progress) , University of Waterloo

Co-Supervisor Thesis/Project Title: Quadratic Gravity in Cosmology and Black Holes

Present Position: Faculty of Mathematics William T. Tutte Postdoctoral Fellow, Department of Applied Mathematics, University of Waterloo

2022/8 - 2024/12 Hassani, Farbod (In Progress) , University of Oslo

Principal Supervisor Thesis/Project Title: Project Supervision: "Simulating dynamical friction with cosmic neutrinos"

Present Position: Postdoctoral Fellow, University of Oslo

2022/1 - 2024/12 Longo, Luis Felipe (In Progress) , Friedrich-Schiller Universitaet, Jena, Germany

Principal Supervisor Thesis/Project Title: Spectrogram Correlated Stacking

Present Position: European Research Council (ERC) Postdoctoral Fellow, Friedrich-Schiller Universitaet, Jena, Germany

2021/9 - 2022/8 Muir, Jessica (Completed) , Perimeter Institute for Theoretical Physics

Principal Supervisor Thesis/Project Title: Project supervision: "Search for flattened non-gaussianity"

Present Position: Postdoctoral Fellow, Perimeter Institute for Theoretical Physics

2021/3 - 2025/4 Mylova, Maria (In Progress) , Perimeter Institute for Theoretical Physics and IPMU, Japan

Principal Supervisor Thesis/Project Title: 1- Non-Gaussian Signatures of a Thermal Big Bang 2- Effective Cuscuton Theory

Present Position: Emmy Noether visiting postdoctoral fellow, Perimeter Institute for Theoretical Physics

2020/10 - 2024/12 Nikakhtar, Farnik (In Progress) , Yale University

Co-Supervisor Thesis/Project Title: Project Supervision: "Action-Angle Space Clustering"

Present Position: Postdoctoral Fellow, Yale University

2020/3 - 2024/3 Dey, Ramit (Completed) , Western University, Ontario, Canada

Co-Supervisor Thesis/Project Title: 1- Spectrogram correlated stacking: A novel time-frequency domain analysis of the Stochastic Gravitational Wave Background 2- Black Hole Echoes

Present Position: Research Associate, Western University, Ontario, Canada

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2020/2 - 2020/6 Shoshani, Barak (Completed) , Perimeter Institute for Theoretical Physics

Principal Supervisor Thesis/Project Title: Bayesian Search for Black Hole Echoes

Present Position: Assistant Professor, Physics Department, Brock University

2019/4 - 2021/3 Oshita, Naritaka (Completed) , Perimeter Institute for Theoretical Physics

Principal Supervisor Thesis/Project Title: Echoes from Quantum Black Holes

Present Position: Assistant Professor, Kyoto University, Yukawa Institute, Kyoto

2019/1 - 2023/12 Abedi, Jahed (Completed) , University of Stavanger, Norway

Principal Supervisor Thesis/Project Title: Search for Black Hole Echoes in Gravitational Wave Observations

Present Position: Assistant Professor, Beijing Institute for Mathematical Sciences and Applications (BIMSA)

2015/9 - 2018/6 Nelson, Elliot (Completed) , Perimeter Institute for Theoretical Physics

Principal Supervisor Thesis/Project Title: Quantum Decoherence during Inflation from Gravitational Nonlinearities

Present Position: Research Scientist, IBM Research

2013/9 - 2017/8 Guariento, Daniel (Completed) , Universidade de Sao Paulo/Perimeter Institute for

Principal Supervisor Theoretical Physics

Thesis/Project Title: Cosmological Black Holes from Self-Gravitating Fields

Present Position: Professor, Conestoga College, Canada

2013/9 - 2014/8 Scrimgeour, Morag (Completed) , University of Waterloo

Principal Supervisor Thesis/Project Title: Mass of Neutrinos from Neutrino Wakes

Present Position: Data Scientist, Square, San Francisco CA

2012/9 - 2015/8 Marsh, David (Completed) , Perimeter Institute for Theoretical Physics

Thesis/Project Title: Structure of Dark Matter Haloes

Present Position: Sofja Kovaleskaja Fellow, University of Goettingen

2012/9 - 2016/8 Kubiznak, David (Completed) , Perimeter Institute for Theoretical Physics

Principal Supervisor Thesis/Project Title: Black Hole Chemistry

Present Position: Senior Fellow, Perimeter Scholar International program

2012/9 - 2016/8 Hinterbichler, Kurt (Completed) , Perimeter Institute for Theoretical Physics

Principal Supervisor Thesis/Project Title: Consistent modifications of Einstein gravity

Present Position: Assistant Professor, Case-Western Reserve University

2012/9 - 2015/8 Kobayashi, Takeshi (Completed) , Perimeter Institute for Theoretical Physics/Canadian

Principal Supervisor Institute for Theoretical Astrophysics

Thesis/Project Title: Primordial Magnetic Fields from the Post-Inflationary Universe

Present Position: Postdoctoral Fellow, ICTP Italy

2011/9 - 2013/8 Lavaux, Guilhem (Completed) , University of Waterloo/Perimeter Institute-CITA National Co-Supervisor Fellow

Thesis/Project Title: A New Light on Missing Baryons

Present Position: CNRS Permanent Position, L'Institut d'Astrophysique de Paris, France

2011/9 - 2012/8 Baghran, Shant (Completed) , University of Waterloo/Perimeter Institute for Theoretical

Principal Supervisor Physics

Thesis/Project Title: Prospects for Detecting Dark Matter Halo Substructure with Pulsar Timing

Present Position: Professor at Sharif University, Sharif University, Tehran, Iran

2010/9 - 2013/8 Zavala, Jesus (Completed) , University of Waterloo - CITA National Fellow

Principal Supervisor Thesis/Project Title: Clustering in the Phase Space of Dark Matter Haloes

Present Position: Professor at U-Iceland, University of Iceland

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2010/9 - 2012/8 Johnson, Matthew (Completed) , Perimeter Institute for Theoretical Physics
Principal Supervisor Thesis/Project Title: Determining the Outcome of Cosmic Bubble Collisions in Full General Relativity
Present Position: Associate Professor at York University, Associate Faculty at Perimeter Institute

2009/9 - 2011/8 Shandera, Sarah (Completed) , Perimeter Institute for Theoretical Physics
Principal Supervisor Thesis/Project Title: Breakdown of Semiclassical Methods in de Sitter Space
Present Position: Associate Professor, Penn-State University

2009/9 - 2013/8 Erickcek, Adrienne (Completed) , Perimeter Institute for Theoretical Physics/Canadian
Principal Supervisor Institute for Theoretical Astrophysics
Thesis/Project Title: Catastrophic Consequences of Kicking the Chameleon
Present Position: Assistant Professor, University of North Carolina-Chapel Hill
Research Associate [n=1]

2021/9 - 2022/5 Mukherjee, Suvodip (Completed) , Perimeter Institute for Theoretical Physics
Principal Supervisor Thesis/Project Title: Spectrogram correlated stacking: A novel time-frequency domain analysis of the Stochastic Gravitational Wave Background
Present Position: Reader, Tata Institute of Fundamental Research, India
Staff Supervision
Number of Visiting Researchers: 1
Event Administration

2020/7 - 2024/12 Member of the Organizing Committee, "Copernicus Weekly Webinar Series" focusing on cosmology and high energy physics <https://indico.cern.ch/event/936284/overview>, Seminar, 2020/7 - 2024/12

2023/3 - 2024/7 Local Organizing Committee (Chair), "50 Years of Horndeski Gravity: Exploring Modified Gravity" University of Waterloo and Perimeter Institute for Theoretical Physics, Conference, 2024/7 - 2024/7

2019/10 - 2024/6 Scientific Advisory Committee member, "[RN50] The Event Horizon and Beyond - Celebrating 50 Years of Narayan" Boston Museum of Science, Boston, MA <https://www.cfa.harvard.edu/events/rn50-celebrating-50-years-narayan>, Conference, 2024/6 - 2024/6

2023/11 - 2024/5 Local Organizing Committee (chair), "McNamara@65: Understanding Feedback in Galaxies and Clusters" University of Waterloo, Conference, 2024/5 - 2024/5

2023/11 - 2024/4 Scientific Organizing Committee Member, "Cosmological Frontiers in Fundamental Physics" (University of Edinburgh), Conference, 2024/4 - 2024/4

2022/10 - 2023/5 Organizing Committee (Chair), "Quantum Spacetime in the Cosmos: From Conception to Reality" Perimeter Institute, Conference, 2023/5 - 2023/5

2020/8 - 2021/5 Scientific Organizing Committee Member, "Cosmological Frontiers in Fundamental Physics" (Virtual at APC, Paris), Conference, 2021/5 - 2021/5

2020/2 - 2020/2 Organizer, "Echoes in Southern Ontario" Perimeter Institute, Workshop, 2020/2 - 2020/2

2019/5 - 2019/11 Co-organizer, Everpresent Lambda: Theory meets Observations, Conference, 2019/11 - 2019/11

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2019/3 - 2019/9 Lead Organizer, "Cosmological Frontiers in Fundamental Physics 2019" Perimeter Institute, Conference, 2019/9 - 2019/9

2019/2 - 2019/7 Chairperson of Parallel Session, 22nd International Conference on General Relativity and Gravitation, Conference, 2019/7 - 2019/7

2017/9 - 2019/4 Chair, Astrophysics Seminar Chair, University of Waterloo, Seminar, 2017/9 - 2019/4

2019/1 - 2019/2 Scientific Organizing Committee, Tehran Meeting on Cosmology at the Crossroads (Virtual), Conference, 2019/2 - 2019/2

2017/9 - 2018/12 Chair, Physics 10 Undergraduate Lectures, University of Waterloo, Course, 2017/9 - 2018/12

2018/1 - 2018/6 Co-organizer, Asymptotic Safety in a Dark Universe, Conference, 2018/6 - 2018/6

2017/5 - 2017/11 Co-organizer, Quantum Black Holes in the Sky?, Conference, 2017/11 - 2017/11

2016/1 - 2016/7 Co-organizer, Philosophy of Cosmology, Conference, 2016/7 - 2016/7

2015/6 - 2015/12 Co-organizer, Mann Fest, Conference, 2015/12 - 2015/12

2015/2 - 2015/8 Co-organizer, Cosmic Flows (and Other Novelties on Large Scales), Conference, 2015/8 - 2015/8

2014/3 - 2014/4 Co-organizer, Implications of BICEP2, Workshop, 2014/4 - 2014/4

2011/5 - 2012/5 Co-organizer, Conformal Nature of the Universe, Conference, 2012/5 - 2012/5

2011/10 - 2012/4 Co-organizer, Theoretical Physics Workshop, Workshop, 2012/4 - 2012/4

2010/9 - 2011/9 Co-organizer, Unravelling Dark Matter 2011, Conference, 2011/9 - 2011/9

2009/12 - 2010/6 Co-organizer, Cosmological Frontiers of Fundamental Physics, Workshop, 2010/6 - 2010/6

2008/9 - 2009/8 Chair, Cosmology Seminar Series, Seminar, 2008/9 - 2009/8

2008/11 - 2009/5 Co-organizer, New Prospects for Solving the Cosmological Constant Problem, Workshop, 2009/5 - 2009/5

2008/5 - 2009/5 Co-organizer, Sunyaev-Zel'dovich Universe and the Future of Cluster Cosmology, Conference, 2009/4 - 2009/5

2008/10 - 2008/12 Co-organizer, Young Researchers Conference, Conference, 2008/12 - 2008/12

2008/4 - 2008/10 Co-organizer, PI/CITA Cosmology Mini-Workshop, Workshop, 2008/10 - 2008/10

2007/12 - 2008/6 Co-organizer, Small Scale Structure of Dark Matter, Workshop, 2008/6 - 2008/6

2008/1 - 2008/3 Co-organizer, Origins and Observations of Primordial Non-Gaussianity, Workshop, 2008/3 - 2008/3

Editorial Activities

2014/1 - 2020/8 Scientific Editor, Research in Astronomy and Astrophysics, Journal

2012/9 - 2013/8 Editorial Board Member, Phys13 News, Newsletter

2010/9 - 2011/8 Editorial Board Member, Phys13 News, Newsletter

Graduate Examination Activities

2024/9 - 2024/9 PhD External Examiner, Danieli, Valentina, Astroparticle Physics, Scuola Inter. Superiore di Studi Avanzati

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2024/9 - 2024/9 PhD External Examiner, Velluci, Vania, Astroparticle Physics, Scuola Inter. Superiore di Studi Avanzati

Community and Volunteer Activities

2009/9 Speaker/Educator, Perimeter Institute for Theoretical Physics

I am a regular contributor to outreach events at Perimeter Institute. My responsibilities include giving lectures, participating in Q&A's, and discussions with primary and high school students, as well as high school teachers.

Knowledge and Technology Translation

2020/2 - 2023/5 Creator, Research Uptake Strategies

Target Stakeholder: General Public

Outcome / Deliverable: Based on our study of North American demographic, mobility, climate, and COVID-19 data, we provided a predictive model for growth and mitigation strategies for each US and Canadian city and county. The models were publicly available in the form of an online portal, during April 2021- May 2023.

References / Citations / Web Sites: Canada: <http://mylocalcovid.uwaterloo.ca>

[decommissioned in May 2023] US: <https://wolfr.am/COVID19Dash>

2020/4 - 2021/10 Scientific Advisor, Consultation Service

Group/Organization/Business Served: Public Broadcasting Company (PBS)

Target Stakeholder: General Public

Outcome / Deliverable: NOVA Universe Revealed: Big Bang, Venture back to the birth of space and time as NOVA probes the mystery of the Big Bang. Premiered: 11/24/21 at 9pm EST Runtime: 53:16, PBS documentary

Evidence of Uptake/Impact: While the specific viewership for this program is not publicly released, PBS is the premier public national broadcaster in the US. Each month, approximately 100 million people watch their local PBS stations.

References / Citations / Web Sites: <https://www.pbs.org/wgbh/nova/video/nova-universe-revealed-big-bang/>

Activity Description: I acted as the scientific advisor for the PBS NOVA documentary: "NOVA Universe Revealed: Big Bang", and helped with creating a scientifically accurate portrait of the Big Bang

International Collaboration Activities

2019/8 - 2029/8 Member of Laser Interferometer Space Antenna (LISA) consortium, Canada

The Laser Interferometer Space Antenna (LISA) is a planned space-based observatory designed to detect and study gravitational waves. As a member of the LISA consortium, I am contributing to the agenda and ideas for the "Fundamental Physics" and "Waveform Modelling" working groups.

Committee Memberships

2024/7 - 2025/7 Committee Member, Seminars and Colloquia Committee, Perimeter Institute for Theoretical Physics

2020/9 - 2024/8 Committee Member, Undergraduate Experience Committee, University of Waterloo
Organize and coordinate events to improve student experience

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2023/9 - 2024/7 Committee Member, Postdoctoral Search Committee, Perimeter Institute for Theoretical Physics

2021/9 - 2023/8 Chair, PhD Student Student Affairs, Perimeter Institute for Theoretical Physics

2013/1 - 2022/7 Committee Member, Early Researcher Award Grant Review Panel, Ministry of Research and Innovation, Government of Ontario

I served on the review panel for most years, during this period.

2019/9 - 2020/8 Committee Member, Emmy Noether Fellowship Search Committee, Perimeter Institute for Theoretical Physics

2019/7 - 2019/10 Chair, "Astrophysics Theory Panel" grant review committee, National Aeronautics and Space Administration (NASA)

Chair of a review panel for Theoretical Astrophysics grants

2014/8 - 2017/8 Chair, Colloquium Committee, Department of Physics and Astronomy, University of Waterloo

2013/9 - 2014/8 Committee Member, Co-op Report Marking Committee, University of Waterloo
Ongoing Membership

2010/9 - 2012/8 Committee Member, Seminar/Colloquium Committee, University of Waterloo

2010/9 - 2011/8 Committee Member, SIN Exam Committee, University of Waterloo

2010/9 - 2011/8 Committee Member, PI/UW Astronomy Faculty Search Committee, University of Waterloo

2008/9 - 2010/8 Committee Member, Postdoctoral Recruitment Committee, Perimeter Institute for Theoretical Physics

Other Memberships

2021/1 Member, Canadian Institute for Theoretical Astrophysics, Incorporated (CITA, Inc)

The Canadian Institute for Theoretical Astrophysics (CITA), located at the University of Toronto, is a leading research centre dedicated to exploring fundamental questions in theoretical astrophysics. CITA's research spans areas such as cosmology, galaxy formation, black holes, high-energy astrophysics, and planetary science. As a member of CITA Inc., I collaborate with other members across the country to advance theoretical astrophysics. As part of this organization, I contribute to recruiting top talent to Canada and mentoring highly qualified personnel, fostering the next generation of astrophysicists in this dynamic field.

2015/1 Ordinary Member, Canadian Astronomical Society (CASCA)

CASCA is a professional society for Canadian astronomers and researchers involved in the fields of astronomy, astrophysics, and related disciplines. The primary goals of CASCA are to promote the advancement of astronomical research, education, and outreach in Canada.

2009/9 - 2014/8 Faculty of Science representative at the Faculty of Math Council, University of Waterloo
Ongoing appointment

Presentations

1. (2025). Quantum Black Holes & the Conundrum of Cosmic Causality. Physics Colloquium, Arizona State University, Phoenix, AZ, United States of America

Main Audience: Researcher

Invited?: Yes

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2. (2025). Battle of the Big Bang: How the Universe Began (and Why It's a Big Mystery!). Physics Literacy for Kids (Zoom talk),
Main Audience: General Public
Invited?: Yes
3. (2025). Copenhagen Survey on Black Holes and Fundamental Physics. Physics Colloquium (virtual),
Tehran, Iran
Main Audience: Researcher
Invited?: Yes, Keynote?: No, Competitive?: No
4. (2025). Quantum Black Holes & the Conundrum of Cosmic Causality. Physics & Astronomy Colloquium,
McMaster University, Hamilton, ON, Canada
Main Audience: Researcher
Invited?: Yes
5. (2025). Quantum Causality & the Crisis in Physics: Black Holes, the Big Bang, and Beyond. Sydney CPPC
virtual seminar, Sydney, Australia
Main Audience: Researcher
Invited?: Yes
6. w/ Stefano Liberati (SISSA). (2024). Quantum Black Holes (concluding panel discussion, and wrap-up).
Black Holes inside and out, Niels Bohr Institute, Copenhagen, Denmark
Main Audience: Researcher
Invited?: Yes, Keynote?: Yes
7. (2024). Precision Quantum Cosmology: Past, Present, and Future. Gravity and Cosmology 2024, Yukawa
Institute for Theoretical Physics, Kyoto University, Kyoto, Japan
Main Audience: Researcher
Invited?: Yes, Keynote?: No
8. (2024). Quantum Spacetime in the Sky: from the horizon to horizon. Heising-Simons Foundation (HSF)
Collaboration on Quantum Gravity and Its Observational Signatures (QuRIOS) Journal Club (virtual),
Waterloo, ON, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
9. (2024). Quantum Black Holes in the Sky. The Event Horizon and Beyond, Celebrating 50 years of Narayan,
Boston, MA, United States of America
Main Audience: Researcher
Invited?: No, Keynote?: No
10. (2024). Quantum Black Holes & the Conundrum of Cosmic Causality. Physics Colloquium, International
Center for Theoretical Physics Asia-Pacific (ICTP-AP), Beijing, China
Main Audience: Researcher
Invited?: Yes
11. (2024). Quantum Spacetime in the Sky: from the horizon to the vacuum. Bottom-Up Cross-Cutting
Workshop “JENAS Initiative: Gravitational Wave Probes of Fundamental Physics” 12–16 Feb 2024
Sapienza University of Rome, Rome, Italy
Main Audience: Researcher
Invited?: Yes, Keynote?: No
12. (2024). Quantum Black Holes & the Conundrum of Cosmic Causality. Physics Colloquium, Institute of High
Energy Physics, Beijing, China
Main Audience: Researcher
Invited?: Yes

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13. (2024). Quantum Spacetime in the Sky: from the horizon to horizon. International Congress of Basic Science 2, Beijing Institute for Mathematical Science and Applications (BIMSA), Beijing, China
Main Audience: Researcher
Invited?: Yes, Keynote?: No
14. (2024). Imaging Quantum Black Holes & (Towards) Optimal Estimators of Stochastic Gravitational Wave Background. The 1st Tsinghua Relativity Workshop, Beijing, China
Main Audience: Researcher
Invited?: Yes, Keynote?: No
15. (2023). The Big Picture: New Vistas in Gravitational Wave Astronomy. Canadian Astronomical Society (CASA) Annual General Meeting, Penticton, BC, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
16. (2023). Quantum Spacetime and the Cosmic Endgame. Physics Colloquium, Simon Fraser University, Burnaby, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: Yes
17. (2023). A Tale of Stars and Sirens. The Past, Present and Future of Astronomical Surveys - A Celebration of the Scientific Careers of Michael Strauss and Robert Lupton, Princeton University, Princeton, NJ, United States of America
Main Audience: Researcher
Invited?: No, Keynote?: No
18. (2023). Quantum Gravity and the Cosmos. Testing Aspects of General Relativity-II (virtual), Waterloo, ON, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
19. (2023). The paradox of human and the cosmos. Ottawa Intellectual Discussions (Virtual), Waterloo, ON, Canada
Main Audience: General Public
Invited?: Yes, Keynote?: Yes
20. (2023). The Endgame: Quantum Spacetime and the Cosmos. Physics Colloquium, University of British Columbia, Vancouver, BC, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: Yes
21. (2023). The Big Picture. DE Shaw Company (virtual seminar), Waterloo, ON, Canada
Main Audience: Decision Maker
Invited?: Yes, Keynote?: No
22. (2023). Quantum Spacetime in the Sky: from Conception to Reality. ISRA 2023: The International Symposium on Recent Developments in Relativistic Astrophysics, Gangtok, India
Main Audience: Researcher
Invited?: Yes, Keynote?: No
23. (2023). Into the Unknown: A Journey on the Dark Side. Virtual Public Lecture for Lord Byng high school Astronomy Club (Vancouver, BC), Waterloo, ON, Canada
Main Audience: General Public
Invited?: Yes, Keynote?: Yes
24. (2023). From Black Holes to the Big Bang: Humanity's Cosmic Paradox. Public Lecture at Kitchener Public Library, Kitchener, ON, Canada

Main Audience: General Public

Invited?: Yes, Keynote?: Yes

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25. (2023). Spectrogram correlated stacking: A novel time-frequency domain analysis of the Stochastic Gravitational Wave Background. Cosmology and Gravitation Seminar, Perimeter Institute, Waterloo, ON, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
26. (2023). Quantum Precision Cosmology. Testing Gravity 2023, Vancouver, BC, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
27. (2023). The War Within: Humanity's Cosmic Paradox. Physics 10 seminar series, University of Waterloo, Waterloo, ON, Canada
Main Audience: General Public
Invited?: Yes, Keynote?: Yes
28. (2022). Einstein vs Einstein. Testing Aspects of General Relativity (Joint Canadian/Indian virtual conference), Waterloo, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: Yes
29. (2022). Einstein vs Einstein: The Great War. Physics Colloquium Institute for Research in Fundamental Sciences (IPM), (remote presentation), Tehran, Iran
Main Audience: Researcher
Invited?: Yes, Keynote?: Yes
30. (2022). Mapping the Big Bang. Remote Public Talk at Farzanegan High School, Tehran (virtual talk in Persian), Tehran, Iran
Main Audience: General Public
Invited?: Yes, Keynote?: Yes
31. (2022). Stimulating Hawking Radiation of Gravitational waves: from theory to observation. New horizons for (no-)horizon physics: from gauge to gravity and back, Galileo Galilei Institute, Florence, Italy
Main Audience: Researcher
Invited?: Yes, Keynote?: No
32. (2022). Einstein vs Einstein: The Great War. Cosmo Coffee Seminar Series, Theory Group, the European Organization for Nuclear Research (CERN) (remote presentation), Geneva, Switzerland
Main Audience: Researcher
Invited?: Yes, Keynote?: Yes
33. (2022). Stimulating Hawking Radiation of Gravitational Waves. Quantum Simulators of Fundamental Physics (virtual seminar series), Waterloo, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: Yes
34. (2022). Quantum Black Holes: from holography to scrambling and gravitational wave astronomy. The 1st International Conference of Holography and its Applications, Damghan University (remote presentation), Damghan, Iran
Main Audience: Researcher
Invited?: Yes, Keynote?: Yes
35. (2022). Fundamental Physics vs. Fundamentalism. History, Philosophy, and Sociology of Cosmology and Astroparticle Physics Conference, Bonn, Germany
Main Audience: Researcher
Invited?: No, Keynote?: No
36. (2021). Cosmology of the Pandemic. Journal-club Galaxies, Institut d'Astrophysique de Paris (remote talk),

Paris, France

Main Audience: Researcher

Invited?: Yes, Keynote?: Yes

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37. (2021). Einstein vs Einstein: The Great War. Gravity Seminar Series, University of Southampton (remote presentation), Southampton, United Kingdom

Main Audience: Researcher

Invited?: Yes, Keynote?: Yes

38. (2021). Black Holes, Unscripted. Copernicus Webinar Series (virtual presentation), Waterloo, Canada

Main Audience: Researcher

Invited?: Yes, Keynote?: Yes

39. (2021). On the origin of the LIGO "mystery" noise and the high energy particle physics desert. Sixteenth Marcel Grossmann Meeting (virtual), Waterloo, ON, Canada

Main Audience: Researcher

Invited?: Yes, Keynote?: No

40. (2021). Physics and Astrophysics with Gravitational Waves. 14th Iranian national conference on Astronomy and Astrophysics, Semnan University (Virtual public talk in Persian), Semnan, Iran

Main Audience: General Public

Invited?: Yes, Keynote?: Yes

41. (2021). Big Bang: Fact vs Fiction. The 28th IPM Physics Spring Conference, Tehran, Iran (remote presentation), Tehran, Iran

Main Audience: Researcher

Invited?: Yes, Keynote?: No

42. (2021). Cosmology and the Pandemic. Virtual conference on Gravity and Cosmology, Shahid Beheshti University (Remote keynote presentation in Persian), Tehran, Iran

Main Audience: General Public

Invited?: Yes, Keynote?: Yes

43. (2021). A New Way to Measure Mass. Streams 21 (virtual conference), Waterloo, Canada

Main Audience: Researcher

Invited?: No, Keynote?: No

44. (2021). Cosmology: Triumphs, Trials, and Tribulations. Physics Colloquium, Tarbiat Modares University (remote presentation), Tehran, Iran

Main Audience: Researcher

Invited?: Yes, Keynote?: Yes

45. (2021). From Cosmos to the Pandemic. 37th Science and Technology Conference, Allameh Helli High School (virtual talk in Persian), Tehran, Iran

Main Audience: General Public

Invited?: Yes, Keynote?: Yes

46. (2021). Into the Unknown. Physics Colloquium, University of Northern Iowa (remote), Cedar Falls, IA, United States of America

Main Audience: General Public

Invited?: Yes, Keynote?: Yes

47. (2021). Black Holes, Unscripted. Black-Hole Microstructure, virtual conference organized by U-Saclay, France (remote presentation), Saclay, France

Main Audience: Researcher

Invited?: Yes, Keynote?: No

48. (2021). From Cosmology to the Pandemic. Sharif University of Technology Alumni Associate Webinar Series (virtual talk in Persian), Waterloo, ON, Canada

Main Audience: Decision Maker

Invited?: Yes, Keynote?: Yes

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49. (2021). On Nothing!. Alternative Gravities and Fundamental Cosmology 2021 (virtual conference, organized by University of Szczecin, Poland), Szczecin, Poland

Main Audience: Researcher

Invited?: Yes, Keynote?: Yes

50. (2021). On the origin of the LIGO "mystery" noise and the high energy particle physics desert. The XXVIII International Conference on Supersymmetry and Unification of Fundamental Interactions (SUSY 2021, virtual), Waterloo, ON, Canada

Main Audience: Researcher

Invited?: Yes, Keynote?: No

51. (2021). From the Cosmos to the Pandemic: A Dive into the Unknown. 26th Special Physics School, Institute for Advanced Studies in Basic Sciences (remote talk in Persian), Zanzan, Iran

Main Audience: General Public

Invited?: Yes, Keynote?: Yes

52. (2020). Probing Physics Beyond the Standard Model with the Gravitational Wave "Mystery" Noise. Quantum Spacetime and the Renormalization Group (organized by CP3 Origins, Denmark, virtual conference), Waterloo, ON, Canada

Main Audience: Researcher

Invited?: Yes, Keynote?: No

53. (2020). Local drivers of growth and containment for COVID-19 pandemic. Seminar Series on the Mathematical Modelling of COVID-19, Fields Institute for Research in Mathematical Sciences, Toronto, Canada

Main Audience: Researcher

Invited?: Yes, Keynote?: No

54. (2020). A New Way to Measure Mass. CANadian Virtual Astronomy Seminar (CANVAS) (virtual presentation), Waterloo, Canada

Main Audience: Researcher

Invited?: No, Keynote?: No

55. (2020). On the Edge of the Abyss. Virtual Seminar, McGill Space Institute, Montreal, QC, Canada

Main Audience: Researcher

Invited?: Yes, Keynote?: Yes

56. (2020). A New Way to Measure Mass. Galactic Dynamics Journal Club, University of Pennsylvania (remote presentation), Philadelphia, PA, United States of America

Main Audience: Researcher

Invited?: Yes, Keynote?: No

57. (2020). Quantum Black Hole Seismology. The 13th International LISA Symposium (virtual conference, pre-recorded talk), Waterloo, Canada

Main Audience: Researcher

Invited?: No, Keynote?: No

58. (2020). Quantum Black Hole Seismology. 23rd meeting in Astronomical Research, Zanzan, Iran (remote presentation), Zanzan, Iran

Main Audience: Researcher

Invited?: Yes, Keynote?: Yes

59. (2020). A novel predictive framework for the novel coronavirus. Centre for Disease Modelling, Canada-China Distinguished Lecture, Canada

Main Audience: Researcher

Invited?: Yes, Keynote?: No

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60. (2020). Quantum Black Holes in the Sky. Quantum Gravity and the Renormalization Group, Odense, Denmark
Main Audience: Researcher
Invited?: Yes, Keynote?: Yes
61. (2020). Managing the COVID-19 Pandemic across Geography and Demography. seminar for the Google COVID-19 data working group (virtual), Waterloo, ON, Canada
Main Audience: Knowledge User
Invited?: Yes, Keynote?: No
62. (2020). On the secrets of black holes, worm holes, white holes, and the firewalls. Remote public talk in Persian for Astronomy (Nojoom) Magazine, Tehran, Iran
Main Audience: General Public
Invited?: Yes, Keynote?: Yes
63. (2020). Quantum Black Hole Seismology. Astrophysics Seminar Series, Bar-Ilan University, Israel (remote presentation), Israel
Main Audience: Researcher
Invited?: Yes, Keynote?: Yes
64. (2020). Quantum Black Hole Seismology. High Energy Physics (HEP) seminar series, school of physics, Institute for Research in Fundamental Sciences (IPM), Tehran (remote presentation), Tehran, Iran
Main Audience: Researcher
Invited?: Yes, Keynote?: Yes
65. (2020). Managing the COVID-19 Pandemic across Geography and Demography. Seminar Series, Astronomy School, Institute for Research in Fundamental Sciences (IPM), Tehran (remote, tehran, Iran
Main Audience: Researcher
Invited?: Yes, Keynote?: Yes
66. (2020). A New Way to Measure Mass. Astrophysics Seminar, Sharif University (remote presentation), Tehran, Iran
Main Audience: Researcher
Invited?: Yes, Keynote?: Yes
67. (2020). The invisible 99 percent. Physics Club (remote public talk in Persian), Tehran, Iran
Main Audience: General Public
Invited?: Yes, Keynote?: Yes
68. (2020). Managing the COVID-19 Pandemic across Geography and Demography. Colloquium, Perimeter Institute, Waterloo, ON, Canada
Main Audience: Knowledge User
Invited?: Yes, Keynote?: No
69. (2020). Hubble Tension and an Everpresent Lambda. 3rd Online Workshop on Quantum Gravity, “Quantum Gravity meets dark energy” (virtual conference), waterloo, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: Yes
70. *Aslanbeigi, S; *Saravani, M. (2019). Dark Matter from Quantum Gravity. Everpresent Lambda: Theory Meets Observations, Perimeter Institute for Theoretical Physics, Waterloo, ON, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
71. (2019). Dark Energy [On Nothing]. CosmoCruise 2019, Venice, Italy
Main Audience: Researcher
Invited?: Yes, Keynote?: No

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72. (2019). A Tale of Big Bang. Milky Way Persian Science Channel, Iran (virtual), Waterloo, ON, Canada
Main Audience: General Public
Invited?: Yes, Keynote?: Yes
73. (2019). Quantum Black Holes in the Sky. Testing Gravity 2019, Vancouver, BC, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: Yes
74. (2019). Cosmological non-Constant (CnC) Problem. 5th Bahamas Advanced Study Institute and Conferences 2019 (BASIC 2019) (remote presentation), Bahamas
Main Audience: Researcher
Invited?: Yes, Keynote?: No
75. (2019). Quantum Black Holes in the Sky. Testing Gravity 2019, Vancouver, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
76. (2019). On Nothing!. Physics Colloquium at Grand Valley State University, Grand Rapids, MI, United States of America
Main Audience: General Public
Invited?: Yes, Keynote?: No
77. (2018). Quantum Black Holes in the Sky: From Quantum Gravity to Astrophysics and Cosmology. Solvay-Perimeter-APC Workshop on "Cosmological Frontiers in Fundamental Physics" - Solvay Institute, Brussels, Belgium
Main Audience: Researcher
Invited?: Yes, Keynote?: No
78. (2018). Quantum Black Holes in the Sky: from Quantum Gravity to Astrophysics and Cosmology. Cosmology and Gravitation Seminar Series, Perimeter Institute, Waterloo, ON, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
79. (2018). Seeing through Simulation Blindspots. Rotman Summer Institute on Philosophy of Cosmology, Goderich, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
80. (2018). Reviving Aether in Einstein's Universe: Trials, Tribulations, and Triumphs. Lecture for undergraduate Cosmology and Gravitation class at Western University, London, ON, Canada
Main Audience: General Public
Invited?: Yes, Keynote?: Yes
81. (2018). Seeing Cosmic Nanostructure: From Plasma to Dark Matter. Seminar - Canadian Institute for Theoretical Astrophysics, University of Toronto, Toronto, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
82. (2018). Quantum Black Holes in the Sky: from Quantum Gravity to Astrophysics and Cosmology. Physics and Astronomy Colloquium, Western University, London, ON, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: Yes
83. (2018). Quantum Black Holes in the Sky: from Quantum Gravity to Astrophysics and Cosmology. Theoretical High Energy Physics Seminar Series, McGill University, Montreal, QC, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: Yes

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84. (2018). Adios, Lorentz! Hola, Kuhn!. Rotman Summer Institute on Philosophy of Cosmology, Goderich, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
85. (2018). Cosmological non-Constant Problem. Asymptotic Safety in a Dark Universe, Perimeter Institute, Waterloo, ON, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
86. (2018). Quantum Black Holes in the Sky: From Quantum Gravity to Astrophysics and Cosmology. HEP Seminar - McGill University, Montreal, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
87. (2018). The Other 99 Percent: A Dark Cosmos. Keynote lecture for high school teachers, Perimeter Institute, Waterloo, ON, Canada
Main Audience: Knowledge User
Invited?: Yes, Keynote?: Yes
88. (2018). Quantum Black Holes in the Sky: from Quantum Gravity to Astrophysics and Cosmology. Cosmology and Gravitation Seminar, Perimeter Institute, Waterloo, ON, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
89. (2017). Echoes from the Abyss. Testing Gravity 2017, Vancouver, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
90. (2017). Reflections on Spacetime. Cosmology and the Future of Spacetime - Western University, London, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
91. (2017). Echoes from the Abyss. Seminar - Department of Applied Mathematics, University of Waterloo, Waterloo, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
92. (2017). Quantum Black Holes in the Sky. Colloquium - University of Kansas, Lawrence, Kansas, United States of America
Main Audience: Researcher
Invited?: Yes, Keynote?: No
93. (2017). Quantum Black Holes in the Sky. Colloquium - Department of Physics, Ohio State University, Columbus, United States of America
Main Audience: Researcher
Invited?: Yes, Keynote?: No
94. (2017). Quantum Black Holes in the Sky - Plenary Lecture. CosPA 2017, Kyoto, Japan
Main Audience: Researcher
Invited?: Yes, Keynote?: No
95. (2017). How to turn Quantum Gravity into Science?. Probing the Spacetime Fabric: From Concepts to Phenomenology - SISSA, Trieste, Italy
Main Audience: Researcher
Invited?: Yes, Keynote?: No

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96. (2017). Echoes from the Abyss. Physics and Astrophysics at the eXtreme (PAX), Amsterdam, Netherlands
Main Audience: Researcher
Invited?: Yes, Keynote?: No
97. (2017). Seeing Dark in the Extremes: From Nanostructure, to Infinity and Beyond. CPARC Colloquium - Queen's University, Kingston, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
98. (2016). The Many Faces of the Cosmological Constant Problem. Experimental Search for Quantum Gravity - Frankfurt Institute for Advanced Studies, Frankfurt, Germany
Main Audience: Researcher
Invited?: Yes, Keynote?: No
99. (2016). Cosmological Non-Constant Problem: Astrophysical Limits of Particle Physics. HET-Discovery Seminar - Niels Bohr Institute, University of Copenhagen, Copenhagen, Denmark
Main Audience: Researcher
Invited?: Yes, Keynote?: No
100. (2016). E.F.T. is Evil!. Unifying Tests of General Relativity - Walter Burke Institute for Theoretical Physics, Caltech, Pasadena, United States of America
Main Audience: Researcher
Invited?: Yes, Keynote?: No
101. (2016). Accretion in Radiative Equipartition. Seminar - Dark Cosmology Centre, University of Copenhagen, Copenhagen, Denmark
Main Audience: Researcher
Invited?: Yes, Keynote?: No
102. (2015). (Real) Cosmology with Quark-Gluon Plasmas. Cosmology/HEP Seminar - Brookhaven National Laboratory, Upton, New York, United States of America
Main Audience: Researcher
Invited?: Yes, Keynote?: No
103. (2015). Pulsar Timing Constraints on Physics Beyond Standard Model. Mann Fest - University of Waterloo, Waterloo, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
104. (2015). Cosmological (Non-)Constant Problem. Testing Gravity 2015, Vancouver, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
105. (2015). First Erlangen Workshop on Quantum Gravity and Cosmology. High Energy Physics Seminar - University of Toronto, Toronto, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
106. (2015). Cosmological Non-Constant Problem: The Case for TeV Scale Quantum Gravity. First Erlangen Workshop on Quantum Gravity and Cosmology, Erlangen, Germany
Main Audience: Researcher
Invited?: Yes, Keynote?: No
107. (2015). Alice in the (Relativistic) Wonderland. Colloquium - Department of Mathematics, Western University, London, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No

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108. (2014). Probing Dark Matter Nanostructure. Mitchell Workshop on Collider and Dark Matter Physics, College Station, United States of America
Main Audience: Researcher
Invited?: Yes, Keynote?: No
109. (2014). From Black Holes to Big Bang, and Back!. Cosmology Seminar - Perimeter Institute for Theoretical Physics, Waterloo, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
110. (2014). Cosmological (Non)-Constant Problem. Large Angle Anomalies: from a posteriori to a priori - Case-Western Reserve University, Cleveland, United States of America
Main Audience: Researcher
Invited?: Yes, Keynote?: No
111. (2014). An Introduction to the Universe - Plenary Talk. Sharif University Alumni Conference, Tehran, Iran
Invited?: Yes, Keynote?: Yes
112. (2014). From Black Holes to Big Bang, and Back!. Colloquium - Department of Physics, Isfahan University of Technology, Isfahan, Iran
Main Audience: Researcher
Invited?: Yes, Keynote?: No
113. (2014). Probing Dark Matter Nanostructure. Cosmology Seminar - Department of Physics, Sharif University of Technology, Tehran, Iran
Main Audience: Researcher
Invited?: Yes, Keynote?: No
114. (2014). Dark Matter Nanostructure. Astrophysics Seminar - Department of Physics and Astronomy, University of Waterloo, Waterloo, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
115. (2014). From Black Holes to Big Bang, and Back!. Invited Colloquium - School of Astronomy, Institute for Research in Fundamental Sciences (IPM), Tehran, Iran
Main Audience: Researcher
Invited?: Yes, Keynote?: No
116. (2014). An Introduction to the Universe, Q&A. Zaferanieh Amateur Observatory, Tehran, Iran
Main Audience: General Public
Invited?: Yes, Keynote?: No
117. (2014). Life after Lorentz: Quantum Mechanics, Gravity, and the Crisis of Falsifiability. Physics Colloquium - University of Tehran, Tehran, Iran
Main Audience: Researcher
Invited?: Yes, Keynote?: No
118. (2013). Missing Baryons: How Much? How Hot? How Fast?. Cross-Correlating Cosmic Fields Conference, Shanghai, China
Main Audience: Researcher
Invited?: Yes, Keynote?: No
119. Gould E. (2013). A Non-Local Reality: Is there a Phase Uncertainty in Quantum Mechanics?. Quantum Foundations Seminar - Perimeter Institute for Theoretical Physics, Waterloo, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No

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120. (2013). Tales of Dark Matter on Small Scales. Colloquium - Kavli Institute for Astronomy and Astrophysics, Beijing, China

Main Audience: Researcher

Invited?: Yes, Keynote?: No

121. (2013). Dynamical Emergence of Universal Horizons During the Formation of Black Holes. LOOPS 13, Waterloo, Canada

Main Audience: Researcher

Invited?: Yes, Keynote?: No

122. (2013). Out of the White Hole: A Holographic Origin for the Big Bang. Cosmology After Planck 2013, Ann Arbor, United States of America

Main Audience: Researcher

Invited?: Yes, Keynote?: No

123. (2013). Where Will Einstein Fail?. CASCA 2013, Vancouver, Canada

Main Audience: Researcher

Invited?: Yes, Keynote?: No

124. (2013). Why is Particle Physics Lorentz-Invariant?. Kavli Workshop - IPMU Focus Week on Gravity and Lorentz Violation, Kashiwa, Japan

Main Audience: Researcher

Invited?: Yes, Keynote?: No

125. (2013). Life after Lorentz: Quantum Mechanics, Gravity, and the Crisis of Falsifiability. Colloquium - University of Wisconsin-Madison, Madison, United States of America

Main Audience: Researcher

Invited?: Yes, Keynote?: No

126. (2013). Tales of Dark Matter on Small Scales. TAMU Workshop on Dark Matter, College Station, United States of America

Main Audience: Researcher

Invited?: Yes, Keynote?: No

127. (2012). Horava-Lifshitz Cosmology and Dark Matter Holes - 2-Lecture Series. Cape Town International Cosmology School 2012, Stellenbosch, South Africa, Republic of

Main Audience: Researcher

Invited?: Yes, Keynote?: No

128. (2012). Hot Big Bang at a Lifshitz Point. CAP Congress 2012, Calgary, Canada

Main Audience: Researcher

Invited?: Yes, Keynote?: No

129. (2012). Reviving Gravity's Aether in Einstein's Universe. Colloquium - York University, Toronto, Canada

Main Audience: Researcher

Invited?: Yes, Keynote?: No

130. (2012). Tales of Dark Matter on Small Scales. Theoretical Physics Workshop - UWO, London, Canada

Main Audience: Researcher

Invited?: Yes, Keynote?: No

131. (2012). Why is Particle Physics Lorentz Invariant?. Experimental Search for Quantum Gravity: The Hard Facts, Waterloo, Canada

Main Audience: Researcher

Invited?: Yes, Keynote?: No

132. (2012). Modern Cosmology. Cosmology & Gravitation Seminar - Perimeter Institute for Theoretical Physics, Waterloo, Canada

Main Audience: Researcher

Invited?: Yes, Keynote?: No

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133. Bonanno A, Brandenberger R, Magueijo J, Nemiroff R, Amelino-Camelia G. (2012). Photons and Quantum Gravity in Astrophysics. Experimental Search for Quantum Gravity 2012, Waterloo, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
134. (2012). Trials and Tribulations of Aether. Conformal Nature of the Universe Conference, Waterloo, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
135. (2011). Where will Einstein Fail? Insights into Gravity and Dark Energy. CITA Invited Seminar - University of Toronto, Toronto, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
136. (2011). Largest and Smallest Dark Matter Clusters, and What We Can Learn from Them. High Energy Theory Seminar - University of Michigan, Ann Arbor, United States of America
Main Audience: Researcher
Invited?: Yes, Keynote?: No
137. (2011). Structure in the Phase Space and Dark Matter Astronomy. Unravelling Dark Matter Conference 2011, Waterloo, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
138. (2011). Where will Einstein Fail? Insights into Gravity and Dark Energy. KICP Colloquium - University of Chicago, Chicago, United States of America
Main Audience: Researcher
Invited?: Yes, Keynote?: No
139. (2011). Aether, Quantum Gravity and Cosmology. Philosophy of Cosmology Workshop - 11th Annual Conference in the Foundations of Physics, London, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
140. (2011). Where will Einstein Fail? Insights into Gravity and Dark Energy. Vainu Bappu Memorial Gold Medal Award Ceremony - Astronomical Society of India, Pune, India
Main Audience: General Public
Invited?: Yes, Keynote?: No
141. (2011). Aether, Quantum Gravity and Cosmology. Cosmological Frontiers in Fundamental Physics 2011, Paris, France
Main Audience: Researcher
Invited?: Yes, Keynote?: No
142. (2011). How can Astrophysical Compact Objects Shed Light on Dark Cosmology?. High Energy Theory Seminar - University of Amsterdam, Amsterdam, Netherlands
Main Audience: Researcher
Invited?: Yes, Keynote?: No
143. (2011). Fine Structure of Dark Matter Phase Space. Dark Matter: Its Origin, Models, and Detection, Albuquerque, United States of America
Main Audience: Researcher
Invited?: Yes, Keynote?: No
144. (2010). Non-Linear Structure Formation in Cosmology - 2-Part Lecture Series. Invited Lectures on Structure Formation - Institute for Research in Fundamental Sciences, Tehran, Iran
Main Audience: Researcher

Invited?: Yes, Keynote?: No

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145. (2010). First Precision Measurement in Quantum Gravity. Cosmology Seminar - Institute for Research in Fundamental Sciences, Tehran, Iran
Main Audience: Researcher
Invited?: Yes, Keynote?: No
146. (2010). Stellar Black Holes and the Thermodynamic Origin of Cosmic Acceleration. Colloquium - Institut d'Astrophysique de Paris, Paris, France
Main Audience: Researcher
Invited?: Yes, Keynote?: No
147. (2010). Cosmological Constant Problem: Rethinking Quantum and Gravity. Laws of Nature: Their Nature and Knowability 2010, Waterloo, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
148. (2010). Cosmology and the End of Physics!. Colloquium - University of Guelph, Guelph, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
149. (2010). Hierarchy in the Phase Space and Dark Matter Astronomy. Astrophysics Seminar - Queen's University, Kingston, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
150. (2009). The End of the Cosmological Constant Problem. Astrophysics Workshop - Interactions in the Dark Sector, Leiden, Netherlands
Main Audience: Researcher
Invited?: Yes, Keynote?: No
151. (2009). Black Hole Horizons and the Origin of Cosmic Acceleration. Holographic Cosmology Workshop 2009, Waterloo, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
152. (2009). Gravitational Aether as a Solution to the CC Problem and more!. New Prospects for Solving the Cosmological Constant Problem 2009, Waterloo, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
153. (2009). Stellar Black Holes and the Origin of Cosmic Acceleration. Cosmology Seminar - KIPAC, Stanford, United States of America
Main Audience: Researcher
Invited?: Yes, Keynote?: No
154. (2009). Hierarchy in the Phase Space and Dark Matter Astronomy. CYGNUS 2009, Cambridge, United States of America
Main Audience: Researcher
Invited?: Yes, Keynote?: No
155. (2009). The End of the Cosmological Constant Problem. Cosmology Seminar - IPMU, Kashiwa, Japan
Main Audience: Researcher
Invited?: Yes, Keynote?: No
156. (2009). Stellar Black Holes and the Origin of Cosmic Acceleration. Theory Canada 5 Conference, Fredericton, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No

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157. (2009). Cuscuton and Solving the Cosmological Constant Problem. High Energy Physics Seminar - Vanderbilt University, Nashville, United States of America
Main Audience: Researcher
Invited?: Yes, Keynote?: No
158. (2009). Cuscuton and Horava-Lifshitz Gravity. Gravity at a Lifshitz Point 2009, Waterloo, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
159. (2009). (Towards) the End of the Cosmological Constant Problem!. Cosmology & Gravitation Seminar - Perimeter Institute for Theoretical Physics, Waterloo, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
160. (2008). Gravitational Aether as a Solution to the Cosmological Constant Problem. 14th International Symposium on Particles, Strings and Cosmology (PASCOS '08), Waterloo, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
161. (2008). Physics Beyond the Horizon. Cosmology and Gravitation Seminar - Perimeter Institute for Theoretical Physics, Waterloo, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
162. (2008). Physics Beyond the Horizon. Physics Colloquium - University of Pennsylvania, Philadelphia, United States of America
Main Audience: Researcher
Invited?: Yes, Keynote?: No
163. (2008). Physics Beyond the Horizon. Colloquium - Perimeter Institute for Theoretical Physics, Waterloo, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
164. (2008). Observational Evidence for Massive Gravity. PI CITA Joint Cosmology Day 2008, Waterloo, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
165. (2008). Missing Thermal Energy of the Universe. Astroparticle Workshop - Open Questions from Clusters of Galaxies, Toulouse, France
Main Audience: Researcher
Invited?: Yes, Keynote?: No
166. (2008). Primordial Non-Gaussianity, Statistics of Collapsed Objects, and the ISW Effect. Origins and Observations of Primordial Non-Gaussianity 2008, Waterloo, Canada
Main Audience: Researcher
Invited?: Yes, Keynote?: No
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Co-Author

Published, University of Chicagao Press,

Refereed?: Yes

Description / Contribution Value: A thrilling exploration of competing cosmological origin stories, comparing new scientific ideas that upend our very notions of space, time, and reality. By most popular accounts, the universe started with a bang some 13.8 billion years ago. But what happened before the Big Bang? And how do we know it happened at all? Here prominent cosmologist Niayesh Afshordi and science communicator Phil Halper offer a tour of the peculiar possibilities: bouncing and cyclic universes, time loops, creations from nothing, multiverses, black hole births, string theories, and holograms. Along the way, they offer both a call for new physics and a riveting story of scientific debate.

Book Chapters

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Supervised Student Publications

1. Aslanbeigi, Siavash AND Saravani, Mehdi

On the Causal Set-Continuum Correspondence (arXiv:1403.6429). JHEP. (2014).

Submitted,

Student Contribution (%): 100

Funding Sources: Natural Sciences and Engineering Research Council of Canada (NSERC)

2. Aslanbeigi, Siavash AND Saravani, Mehdi

Generalized Causal Set d'Alembertians (arXiv:1403.1622). JHEP. (2014). (1406): 024.

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4. Aslanbeigi, Siavash

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Published,

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Magazine Entries

1. Afshordi N, Mann R, Pourhasan R*. The Black Hole That Birthed the Big Bang. Scientific American. 311(2)

Published,

Number of Contributors: 3

Description / Contribution Value: The article, which was featured on the cover of this issue, describes one of our most innovative ideas to a popular audience.

Online Resources

1. Co-Author. Zavala, J, Afshordi, N. P2SAD: Particle Phase Space Average Density (public code). (2014).

Number of Contributors: 2

Description / Contribution Value: P2SAD computes the Particle Phase Space Average Density (P2SAD) in galactic haloes. The model for the calculation is based on the stable clustering hypothesis in phase space, the spherical collapse model, and tidal disruption of substructures. The multiscale prediction for P2SAD computed by this IDL code can be used to estimate signals sensitive to the small scale structure of dark matter distributions (e.g. dark matter annihilation). The code computes P2SAD averaged over the whole virialized region of a Milky-Way-size halo at redshift zero.

Conference Publications

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Published

Refereed?: No, Invited?: Yes

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Paper

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Complete talks and outreach materials inventory

This inventory contains 570 source decks and slide exports from the local *My Talks* archive as catalogued on 2026-07-09. Files remain in the maintained archive; the web collection is at nafshordi.com/talks.

Abyss — 5 items

- RN50 talk template 2024; Keynote
- RN50 talk template 2024; PowerPoint
- Abyss MSI McGill 2020 2020; Keynote
- Testing Aspects of General Relativity-II Apr 2023 2023; Keynote
- Niayesh Ramesh June 2024 2024; Keynote

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- CC philosophy UWO may 11 2011; Keynote
- Gueph Colloquium Oct 19 2010; PowerPoint
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- KITPC Mar 09 2009; PowerPoint
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- PI CC May 09 2009; PowerPoint
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- PI theory canada Jun 09 2009; PowerPoint
- QG measurement ipm aug 10 2010; PowerPoint
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- rustbelt buffalo 2011 Jan 15 2011; PowerPoint
- Aether PI May 2012 2012; Keynote
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AiRE_disk — 8 items

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• NanoGravo+LIGO	2020; Keynote
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- Echo COVID 2 2020; PDF
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- Einstein vs Einstein 2021; PDF
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• Echoes CosPA2017	2017; PDF
• Echoes CosPA2017	2017; Keynote
• Echoes KU 2017	2017; Keynote
• Echoes OSU 2017	2017; Keynote
• Echoes OSU 2017	2017; PDF
• Echoes PAX 2017	2017; Keynote
• Echoes PI 2017	2017; PDF
• Echoes PI 2017	2017; Keynote
• Echoes UW 2017	2017; Keynote
• Echoes Vancouver 2017	2017; PDF
• Echoes Vancouver 2017	2017; Keynote
• Echoes McGill Apr 2018	2018; Keynote
• Echoes PI Feb 2018	2018; Keynote
• Echoes Solvay May 2018	2018; Keynote
• Echoes Western Mar 2018	2018; Keynote
• Echoes TG19 Jan 2019	2019; PDF
• Echoes TG19 Jan 2019	2019; Keynote
• Echoes status update PI Jan 2020	2020; Keynote
• LISA echoes 2020	2020; PDF
• LISA echoes 2020	2020; Keynote
• Stimulating Hawking (GGI 2022)	2022; Keynote
• Stimulating Hawking (GGI 2022)	2022; PDF
• Stimulating Hawking (QSimFP 2022)	2022; Keynote

EFT — 3 items

• EFT CalTech 16	2016; Keynote
• EFT CalTech 16 afshordi	2016; PDF
• EFT CalTech 16 afshordi	2016; Keynote

Endgame_colloquia — 6 items

• SFU Nov 2023	2023; Keynote
• SFU Nov 2023	2023; PDF
• Sikkim Bani Dec 2023	2023; Keynote
• UBC Jan 2023	2023; Keynote
• Rome Feb 2024	2024; Keynote
• Rome Feb 2024	2024; PDF

FPO — 2 items

• FPO	2004; PowerPoint
• FPO	2004; Web slides

Gaia — 7 items

- Gaia action PI 2019 2019; Keynote
- Gaia action Canvas 2020 2020; Keynote
- Gaia action Penn 2020 2020; Keynote
- Gaia action Sharif 2020 2020; Keynote
- Gaia action Sharif 2020 2020; PDF
- Hubble Trouble QG Jun 2020 2020; PDF
- Gaia action streams21 2021 2021; Keynote

General archive — 44 items

- Chairing Seminars 2024; Keynote
- colloquia at UW 2022; Keynote
- colloquia at UW 2014; PDF
- concluding remarks Horndeski 2024; Keynote
- DM ball 2011; Keynote
- GWPI poster 2010; PowerPoint
- GWPI poster 2010; PDF
- Iranian Caldendar 2009; PowerPoint
- Krishan 2023; Keynote
- Kruskal 2023; Keynote
- Matter 2025; Keynote
- mitseminar.pdf 2008; PDF
- ng 2010; PowerPoint
- ng 2010; PowerPoint
- Tips for Talks 2014; PDF
- Tips for Talks 2022; Keynote
- UW jamboree 2009; PowerPoint
- UW jamboree 16 2016; Keynote
- UW jamboree 16 2016; PDF
- Welcome 2023; PDF
- Welcome 2023; Keynote
- Welcome cosmic flows 2015; Keynote
- Gong Show Buffalo 2011 2011; PowerPoint
- Gong Show Buffalo 2011 2011; PDF
- LathamSACslides2011v6 2011; PowerPoint
- Jamboree Oct 2012 2012; PDF
- QG conference Oct 2012 2012; Keynote
- Jamboree PI 2013 2013; Keynote
- Jamboree PI 2013 2013; PDF
- SuperJedi galaxies 2013 2013; PDF
- SuperJedi galaxies 2013 2013; Keynote
- Jamboree UW 2014 2014; PDF
- Jamboree UW 2014 2014; Keynote
- wrapup frontiers 2019 2019; Keynote
- wrapup frontiers 2019 2019; PDF
- causal set conference 2020 2020; PDF
- causal set conference 2020 2020; Keynote
- For Brian 2020 2020; Keynote
- Jamboree 2020 2020; PDF
- Jamboree 2020 2020; Keynote

- Niayesh David 2021 2021; Keynote
- Niayesh David 2021 2021; PDF
- Niayesh WCA Jamboree 2023 2023; Keynote
- Niayesh WCA Jamboree 2023 2023; PDF

Gravity, the ultimate cosmological frontier — 1 items

- gravity 2011; PowerPoint

HL_cosmology — 1 items

- hot big bang 2012; Keynote

ISW_anomaly — 2 items

- case low l 2010 2010; PDF
- case low l 2010 2010; PowerPoint

Jamboree_2018 — 43 items

- AA welcome 2018; PDF
- Afshordi 2.29.45 AM 2018; PDF
- Altamirano 2018; PDF
- Babyk press 2018; PDF
- Balogh 2018; PDF
- Binder2 2018; PDF
- Binder3 2018; PDF
- Bobotsis - Astro Jamboree Slide 2018; PDF
- Broderick 2018; PDF
- combined 2018; PDF
- ConnorMartz 2018; PDF
- Conrad Jamboree Winter2018 2018; PDF
- Drakos AstroJamboree 2018; PDF
- FichJamboree2018 2018; PDF
- Georgiev Astro Jamboree 2018 2018; PDF
- Hudson Jamboree 2018; PDF
- Jamboree2018 2018; Keynote
- Jamboree2018 2018; PDF
- Jamboree2018 1.24.46 AM 2018; PDF
- Jeter 2018; PDF
- Karami 2018; PDF
- mann 2018; PDF
- McNamara 2018; PDF
- morawetz slide 2018; PowerPoint
- morawetz slide 2018; PDF
- Ni 2018; PDF
- Okoli 2018; PDF
- Percival 31-1-18 2018; PDF
- Preciado 2018; PDF
- PU 2018; PDF
- Reeves 2018; PDF
- Robbins - Astro Jamboree - January 31, 2018 2018; PDF
- Rufus 2018; PowerPoint

• SpitzerJamboree	2018; PDF
• Tamhane	2018; PDF
• Taylor	2018; PDF
• Tiede a	2018; PDF
• Vantygghem	2018; PDF
• Wang	2018; PDF
• webb jamboree slide	2018; PDF
• Xi	2018; PDF
• Yang	2018; PDF
• ZZ pizza	2018; PDF

Jamboree_2019 — 40 items

• AAA	2019; PDF
• Amber Hollinger	2019; PDF
• Andrej Obuljen	2019; PDF
• Andrew Reeves	2019; PDF
• Avery Broderick	2019; PDF
• Avery Broderick	2019; Keynote
• Bailey Robison	2019; PDF
• Boris Georgiev	2019; PDF
• Boris Georgiev	2019; Keynote
• Brian McNamara	2019; PDF
• Britt Jetter	2019; PDF
• Chengyu Xi	2019; PDF
• Connor Martz	2019; PDF
• Dustin Lang	2019; PDF
• Faizan Mohammad	2019; PDF
• Ghazal Geshnizjani	2019; PDF
• Isaac Spitzer	2019; PDF
• Jamboree 2019	2019; PDF
• Jamboree2019	2019; Keynote
• James Taylor	2019; PDF
• Karen Mcnab	2019; PDF
• Krishan Saraswat	2019; PDF
• Kristi Webb	2019; PDF
• Matthew Robbins	2019; PDF
• mcnab-jamboree-2018	2019; PowerPoint
• Michael Balogh	2019; PDF
• Michael Chapman	2019; PDF
• Mike Fich	2019; PDF
• Mike Hudson	2019; PDF
• Niayesh Afshordi	2019; PDF
• Nicole Drakos	2019; PDF
• Prathamesh Tamhane	2019; PDF
• Qingwen Wang	2019; PDF
• Robb Mann	2019; PDF
• Robb Mann	2019; PowerPoint
• Supranta Bourah	2019; PDF
• Tianyi Yang	2019; PDF
• Tianyi Yang	2019; Keynote
• Will Percival	2019; PDF
• ZZZ	2019; PDF

Job Interviews — 2 items

- Dark Cosmology 2011; Keynote
- QMUL interview 2023 2023; Keynote

Leiden_Apr_08 — 1 items

- neutrinos and DE 2009; PowerPoint

London_conference_07 — 3 items

- backreaction london 2007; PowerPoint
- szmap london 2007; PDF
- szmap london 2007; PowerPoint

massive gravity — 3 items

- massive 2008; PDF
- massive 2008; PowerPoint
- massive 2008; PowerPoint

massive neutrinos — 3 items

- neutrino florence 2016; PDF
- neutrino florence 2016; Keynote
- neutrino PI 26 2026; Keynote

Michigan — 2 items

- Journey Michigan 2007; PowerPoint
- Journey Michigan 2007; PowerPoint

missing baryons — 2 items

- missing shanghai 2013 2013; PDF
- missing shanghai 2013 2013; Keynote

New Mexico — 3 items

- Dark Side 2007; PDF
- Dark Side 2007; PowerPoint
- Dark Side 2007; PowerPoint

non-gaussianity — 2 items

- non-gaussianity 2008; PowerPoint
- non-gaussianity sharif 2008; PowerPoint

Nothing — 6 items

- GVSU colloquium 2019; PDF
- GVSU colloquium 2019; Keynote
- Hubble Trouble 2019; Keynote
- Hubble Trouble QG Jun 2020 2020; Keynote
- Hubble Trouble QG Jun 2020 2020; PDF
- AltecsmoFun 2021 2021; Keynote

Oxford_interview — 3 items

- oxford interview 2007; PowerPoint
- oxford talk 2007; PowerPoint
- oxford talk 2007; PowerPoint

PDT — 2 items

- PDT Echoes 2020; Keynote
- PDT Echoes 2020; PDF

Penn_job_tal_08 — 1 items

- beyond horizon Penn 2008; PowerPoint

Philosophy — 2 items

- Fundamentalism 2022; Keynote
- Fundamentalism 2022; PDF

PI_job_talk_08 — 1 items

- beyond horizon PI 2008; PowerPoint

Pitt — 2 items

- Dark Side 2007; PowerPoint
- Research Plan 2007; PowerPoint

Poster_Maryland — 1 items

- ISW 2004; PowerPoint

Power Spectra — 1 items

- Picture McGill 2019; Keynote

Precision Quantum Cosmology — 8 items

- Testing Gravity Vancouver Jan 2023 2023; Keynote
- ICBS Beijing 2024 2024; Keynote
- ICBS Beijing 2024 2024; PDF
- ICBS July 2024 2024; Keynote
- ICBS July 2024 2024; PDF
- Kyoto Feb 2024 2024; PDF
- Kyoto Feb 2024 2024; Keynote
- QuRIOS Apr 2024 2024; Keynote

Primordial Black Holes as Dark Matter — 2 items

- PBH 2003; Web slides
- Primordial Black Holes as Dark Matter 2003; PowerPoint

princeton_apr07 — 2 items

- wmapII cosmology 2007; PowerPoint
- wmapII cosmology 2007; PowerPoint

Proof! — 2 items

- Proof 2015; Keynote
- Proof 2015; PDF

PSI — 9 items

- PSI presentation 2009; PowerPoint
- PSI presentation 2009; PDF
- PSI 2019 2019; Keynote
- PSI 2021 2021; Keynote
- PSI 2022 2022; Keynote
- PSI 2023 2023; Keynote
- PSI 2023 1 2023; Keynote
- PSI 2024 2024; Keynote
- PSI 2025 2025; Keynote

public talk — 28 items

- Battle of the Big Bang Secular AZ 2025; Keynote
- dark cosmos 2009; PowerPoint
- Dragons 2.29.43 AM 2018; Keynote
- Dragons 2.29.44 AM 2018; PDF
- Geometry 2012; PowerPoint
- Hologram 2017; Keynote
- Hologram 2017; PDF
- Paradox Ottawa Zoom 2023; Keynote
- PSI 1 2009; PowerPoint
- PSI 1 2009; PDF
- PSI 1 2010; PowerPoint
- Rahe Shiri 2025; Keynote
- Rahe Shiri 2019; PDF
- The Other 99 Percent 2018; PDF
- What Banged 2017; PDF
- What Banged 2025; Keynote
- Geometry EinsteinPlus July 2012 2012; PowerPoint
- Horiozns 2014 Phys10 2014; Keynote
- introduction universe sharif 2014 2014; PowerPoint
- Perseids welcome 2014 2014; Keynote
- Soap 2015 2015; Keynote
- The Other 99 Percent PI Jul 2018 2018; Keynote
- e-Tek 2021 2021; Keynote
- Lord Byng Astro Club, May 2023 2023; Keynote
- Paradox Kitchenre Public Library 2023 2023; Keynote
- Paradox Phys10 2023 2023; Keynote
- Battle of the Big Bang Howard 2026 2026; Keynote
- Battle of the Big Bang KPL 2026 2026; Keynote

Qeshm_School_2005 — 3 items

- Large Scale Structure 2005; PowerPoint
- Secondaries Observations 2005; PowerPoint
- Secondaries Theory 2005; PowerPoint

QPOs — 2 items

- QPOs 2003; Web slides
- QPOs 2003; PowerPoint

Quadratic Gravity — 12 items

- Hangzhou panel Afshordi 2026; Keynote
- Hangzhou panel Afshordi 2026; PDF
- QQG SAB 2026; Keynote
- UV-Universe-QG-DIAS 2025; PowerPoint
- UV-Universe-QG-DIAS 2025; Keynote
- UV-Universe-QG-Imperial 2025; Keynote
- UV-Universe-QG-Kings 2025; Keynote
- QG Wild MannFest 2025 2025; Keynote
- QG Wild MannFest 2025 2025; PowerPoint
- QG Wild RRI 2025 2025; Keynote
- QG Wild Hangzhou 2026 2026; Keynote
- QG Wild Hangzhou 2026 2026; PDF

Quantum Black Holes — 7 items

- QBH IHEP Beijing 2024 2024; Keynote
- QBH UCAS Beijing 2024 2024; Keynote
- Afshordi QBH Simons 2025 2025; Keynote
- QBH ASU 2025 2025; Keynote
- QBH McMaster 2025 2025; Keynote
- QBH Simons 2025 2025; Keynote
- QBH Sydney 2025 2025; Keynote

Reflections — 4 items

- Fabric SISSA 2017 2017; PDF
- Fabric SISSA 2017 2017; Keynote
- Refelction Western 2017 2017; Keynote
- Refelction Western 2017 2017; PDF

Sharif_07 — 1 items

- imprecision cosmology 2007; PowerPoint

Strauss:Lupton Fest — 2 items

- Strauss:Lupton Fest 2023; PDF
- Strauss:Lupton Fest 2023; Keynote

syracuse — 3 items

- beyond horizon Syracuse 2008; PowerPoint
- beyond horizon Syracuse 2008; PowerPoint
- szplane 2008; PowerPoint

SZ_map_Texas — 4 items

- slacconf two 2004; PDF
- Texas Poster 2004; PowerPoint
- Texas Talk 2004; PDF
- Texas Talk 2004; PowerPoint

SZ_sep_chicago — 1 items

- SZ 2003; PowerPoint

szmap — 9 items

- digging 2007; PowerPoint
- szmap 2004; PDF
- szmap 2004; PowerPoint
- szmap aspen 2007; PowerPoint
- szmap benasque 2006; PowerPoint
- szmap fermilab 2006; PowerPoint
- szmap irvine 2006; PowerPoint
- szmap penn state 2007; PowerPoint
- szmap pi 2006; PowerPoint

The Big Picture — 5 items

- Afshordi N talk mac 2023; PDF
- Afshordi N talk mac 2023; Keynote
- The Big Picture @DEShaw 2023; Keynote
- The Big Picture @UCL 2025; Keynote
- The Big Picture @CASCA 2023 2023; Keynote

The Nature of Scientific Knowledge — 5 items

- Knowledge 2018; Keynote
- Knowledge 2019 2019; PDF
- Knowledge 2019 2019; Keynote
- Knowledge 2020 2020; Keynote
- Knowledge 2020 2020; PDF

Thin Accretion Disks — 2 items

- Thin Accretion Disks 2002; PowerPoint
- Thin Accretion Disks 2005; PDF

toulouse — 2 items

- szplane toulouse 2008; PDF
- szplane toulouse 2008; PowerPoint

UNC — 1 items

- Dark Side 2007; PowerPoint

UW luster workshop July 08 — 1 items

- SZ paradox 2008; PowerPoint

UWC chair 2012 — 2 items

- UWC chair presentation 2012; Keynote
- UWC chair presentation 2012; PDF

What I did during the last 12 months — 2 items

- PBH 2003; PDF
- What I did during the last 12 months 2003; PowerPoint

What I did during the last 17 months — 2 items

- What I did during the last 17 months 2002; PowerPoint
- 0202409 2024; PDF

what is wrong with cosmology — 1 items

- what is wrong with cosmology 2008; PowerPoint

white hole — 3 items

- white hole miami dec 2013 2013; Keynote
- white hole miami dec 2013 2013; PDF
- white hole michigan sep 2013 2013; Keynote

York — 1 items

- Dark Side 2007; PowerPoint